TeaTech

Digital Coupon Management System for Hyper-Local Vending

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Declaration

In accordance with the requirements for the degree of B.Tech. Program in Computer Science Engineering in the Faculty of Engineering and Technology, I present this project report on "**TeaTech: Digital Coupon Management System for Hyper-Local Vending**".
I declare that the work presented in the report is my own work except as acknowledged in the text and foot notes.

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Certificate From Guide

This is to certify that **Sidharth Deora** Roll No:-22ECLCS025 has submitted the project report entitled "**TeaTech Digital Coupon Management System for Hyper-Local Vending**" in partial fulfillment for the award of the degree of bachelor of Technology in **Computer Science Engineering**. The report is up to my satisfaction and as per the format prescribed for the writing of the report. His work is approved for presentation.

Date :-06-12-2025 **Signature of Guide**

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CHAPTER 1: INTRODUCTION

1. Project Overview and Rationale

The increasing digitization of corporate operations has transformed how organizations manage internal workflows, employee services, and micro-transactions. However, despite advancements in enterprise software systems, many hyper-local services operating within corporate environments—such as tea and coffee vendors—continue to rely on manual, paper-based mechanisms for daily transactions. One such long-standing mechanism is the distribution and redemption of physical tea or coffee coupons, an approach still widely prevalent in large office complexes, IT parks, and industrial campuses. Although simple in concept, this system introduces numerous inefficiencies that become increasingly apparent as transaction volumes scale. TeaTech emerges as a targeted response to this overlooked yet persistent operational challenge.

TeaTech is a web-based digital coupon management platform designed to replace the traditional paper coupon system with a secure, real-time, and transparent digital alternative. The platform allows employees to maintain a digital coupon balance and redeem it seamlessly at local beverage counters, while vendors can process coupon deductions through an intuitive interface. The system centralizes coupon-related interactions and stores all corresponding records in a tamper-proof database using Firebase Firestore. By doing so, TeaTech not only eliminates the fragility and unreliability associated with paper coupons but also enhances trust between vendors and employees through verifiable audit trails and controlled transaction workflows.

The rationale behind TeaTech is rooted in the recognition that micro-scale financial systems—though minimal in individual transaction value—are collectively critical for maintaining operational harmony in corporate spaces. In environments where hundreds or thousands of beverage transactions occur daily, vendors and employees often face disagreements regarding coupon counts, untracked deductions, and misplaced coupon books. These issues may seem inconsequential at a surface level, yet they contribute to recurring friction, wasted time, and financial ambiguities. TeaTech directly addresses these pain points by enabling structured accountability and reducing the cognitive load associated with manual tracking.

Another significant motivation for developing TeaTech lies in its emphasis on **zero payment friction**, a design philosophy that avoids the complexities of integrating digital payment gateways such as UPI or card-based systems. Many hyper-local vendors still operate primarily on cash-flow models and are reluctant to adopt systems that require digital settlement or regulatory compliance. By digitizing only the redemption layer, TeaTech retains the simplicity of the vendor's existing financial processes while introducing the benefits of automation, transparency, and traceability. This makes the system both culturally compatible and immediately adoptable, particularly in semi-formal or small-business settings where technological interventions often fail due to overcomplexity.

Furthermore, TeaTech introduces a unique **two-step confirmation mechanism** that significantly enhances transaction reliability. In typical digital vending systems, deductions occur instantly upon a

vendor’s request. However, in high-traffic settings where multiple users interact with a single vendor simultaneously, instant deductions can lead to errors or unintended transactions. TeaTech’s confirmation workflow ensures that a deduction is only executed when the user explicitly approves it, thus establishing a shared verification step that removes ambiguity. This design reflects a practical understanding of the interpersonal dynamics at corporate vending points, where communication lapses or rush-hour scenarios frequently cause accidental coupon exchanges.

From a broader perspective, the project rationale extends into the domain of operational digitization within corporate ecosystems. With companies increasingly investing in digital transformation to streamline both core and auxiliary processes, TeaTech aligns with contemporary organizational goals. It demonstrates how even modest, non-critical workflows can benefit significantly from structured digitization. The project also showcases the potential for scalable, low-infrastructure web solutions powered by real-time databases to resolve localized yet persistent challenges without demanding extensive hardware or software installations.

Lastly, TeaTech’s development was informed by real-world observations gathered during a corporate internship environment, where the shortcomings of paper-based coupon systems were directly experienced. These observations revealed patterns of discrepancy, user frustration, and vendor uncertainty that validated the need for a digital intervention. The prototype’s successful deployment in this environment further reinforced the project's relevance and feasibility, demonstrating that the behavior of stakeholders naturally aligns with the features offered by TeaTech. Collectively, these factors form the foundational rationale for undertaking this project and underscore its significance in enhancing day-to-day operational interactions within corporate spaces.

1.2. Problem Identification

1.2.1. High Error Rate (Accounting Disputes)

Physical coupon-based vending systems, although simple in principle, introduce a remarkably high rate of accounting errors as the volume of transactions increases. In corporate settings where hundreds of employees visit a beverage vendor multiple times a day, the cumulative handling of coupons becomes prone to inconsistencies. These inconsistencies typically arise from manual counting, miscommunication, hurried exchanges during peak hours, and the absence of a standardized method for recording each transaction. As a result, both employees and vendors frequently encounter disputes regarding the number of coupons redeemed versus the number remaining, creating an atmosphere of mistrust and operational inefficiency.

A significant contributor to these disputes is the **lack of real-time synchronization** between coupon issuance and coupon usage. Employees often carry coupon booklets for several days or weeks, and vendors rely on manual logs—or in many cases, no logs at all—to record deductions. Over time, discrepancies naturally accumulate due to human memory limitations, unintentional oversight, or environmental factors such as misplaced coupons. When employees believe they have a certain number of coupons remaining while the vendor claims otherwise, verifying the correct count becomes nearly

impossible. The absence of a unified transactional reference is the root cause of such recurrent disagreements.

Another major driver of transaction errors is the **high-pressure environment in which these exchanges occur**. Vendors typically operate with long queues during break hours, leading them to process multiple customers in rapid succession. Under such circumstances, errors like marking extra coupon cuts, failing to mark an actual deduction, or confusing coupons between two consecutive customers are common. While individually small, these errors accumulate significantly over time. For users, this manifests as sudden unexplained drops in coupon balance; for vendors, it appears as undercounting or overcounting of redeemed coupons at day-end or month-end reconciliations.

The variability of human communication further complicates this scenario. Employees may assume a deduction has occurred while the vendor forgets to record it, or the vendor may mistakenly think a deduction was authorized when it was not. As interactions happen quickly and without verification, even slight misinterpretations—such as nodding gestures, casual confirmations, or background noise—can lead to inaccurate accounting. In many cases, neither party has malicious intent; the system simply lacks a dependable mechanism for cross-validation.

From the vendor's perspective, high error rates translate directly into **financial discrepancies**. Vendors operate on thin margins, and even small mismatches in coupon counts can affect their daily revenue. Many vendors attempt to maintain handwritten logs to mitigate errors, but these logs are vulnerable to incomplete entries, inconsistent formats, and environmental wear-and-tear. Additionally, the task of managing logs itself slows down service during busy hours, increasing the likelihood of both operational delays and accounting mistakes.

For employees, accounting disputes lead to **loss of trust and reduced convenience**. When coupon balances appear to fluctuate without explanation, employees become hesitant to rely on the system. They may begin to keep informal counts themselves, further increasing the chances of mismatch due to parallel tracking. Moreover, disputes—however minor—create uncomfortable vendor–employee interactions that disrupt the smooth functioning of the workplace environment. This friction becomes more pronounced in larger organizations, where the vendor may handle hundreds of customer interactions daily and personalized recall is impossible.

In summary, the high error rate inherent in paper coupon systems stems from **manual operations**, **lack of verification**, **environmental pressures**, and **absence of institutional recordkeeping**. These factors combine to create a workflow that is fundamentally error-prone, especially at scale. Identifying this issue is central to understanding why a digital, verifiable, and user-involved redemption process—such as the one introduced by TeaTech—is not merely beneficial but essential for modern corporate vending environments.

1.2.2. Inventory Loss

Inventory loss is an inherent and inevitable weakness of physical coupon-based systems. In the context of corporate tea and coffee vending, "inventory" refers not only to the physical stock maintained by the vendor but also to the paper coupons held by employees. Both forms of inventory are vulnerable to loss,

misplacement, and damage, leading to inconsistencies that neither party can reliably resolve. This repeated pattern of coupon disappearance or deterioration significantly compromises the functional integrity of the traditional system and contributes to operational inefficiencies for both employees and vendors.

One of the primary causes of inventory loss is the **physical fragility of paper coupons**. Coupons are often small, easy to misplace, and susceptible to tearing, moisture damage, or general wear from daily handling. Employees typically carry coupons in pockets, bags, wallets, or notebooks, where they may mix with other items or fall out unnoticed. When an employee attempts to redeem coupons later, the shortage is usually discovered too late, with no mechanism available to verify where or when the coupons were lost. This creates a direct financial loss for the employee, especially in bulk-purchase scenarios where multiple coupons are bought in advance.

Vendors face a parallel challenge with **coupon accumulation and storage**. Over the course of a day, vendors collect stacks of paper coupons in order to keep track of how many beverages they have served. Given the fast-paced nature of their work environment, these coupons are rarely sorted, counted, or stored systematically during business hours. As a result, vendors often end up with misplaced or mixed bundles of coupons. Simple events—like a stack falling, being blown by wind, or getting wet—can result in immediate and irreversible loss. For small-scale vendors who depend on accurate coupon tallies for their daily earnings, even minor coupon loss can have a significant financial impact.

Beyond physical fragility, another contributing factor to inventory loss is **lack of traceability**. Paper coupons do not contain identifiers that link them to specific transactions or users. When a coupon goes missing, neither the employee nor the vendor can establish evidence of the coupon's previous state. In more formal systems, traceable assets (such as digital tokens or barcoded vouchers) help maintain accountability. In contrast, paper coupons lack any inherent tracking mechanism, leading to inventory gaps that cannot be reconciled retrospectively. The inability to track coupons from issuance to redemption ensures that any loss—whether accidental or unnoticed—remains unaccounted for.

Inventory loss also increases during **high-traffic periods**. Break times, shift changes, and lunchtime rushes are particularly chaotic for both vendors and employees. In these moments, employees may hurriedly flip through coupon booklets, accidentally tear off extra coupons, or drop loose coupons while reaching counters. Vendors, on the other hand, may inadvertently misplace coupons while serving multiple customers consecutively. These peak-hour interactions amplify the weaknesses of a paper-based system, as neither party has adequate time to verify coupon counts on the spot.

Environmental conditions within corporate premises further contribute to coupon loss. Tea and coffee stalls are often located in semi-open or high-movement areas where wind, spilled beverages, and crowded surroundings increase the likelihood of accidental coupon damage or loss. Vendors often operate on small countertops or shared spaces, leaving little room for secure coupon storage. Over time, these environmental influences accumulate, adding yet another layer of unpredictability to the already unstable nature of physical coupon management.

The psychological dimension of inventory loss is also noteworthy. Employees who lose coupons may feel frustration, guilt, or reluctance to report the loss, while vendors who misplace collected coupons may

experience anxiety over daily earnings reconciliation. These emotional undertones can negatively impact the relationship between vendors and customers, fostering a sense of uncertainty even when both parties act in good faith. When such incidents recur, employees may develop mistrust toward the coupon system as a whole, perceiving it as unreliable or prone to failure.

From the organization's perspective, inventory loss indirectly affects workplace efficiency. When employees must repurchase lost coupons or engage in disputes over missing ones, it disrupts routine workflows and consumes unnecessary time. Vendors may also spend additional time recounting or verifying their coupon stacks, which slows service and reduces overall satisfaction. Over long periods, these small inefficiencies accumulate into a measurable operational burden that organizations may overlook due to the informal nature of the transaction system.

In essence, inventory loss highlights the inherent limitations of physical artifacts in fast-paced corporate vending environments. The absence of traceability, structural fragility, environmental susceptibility, and lack of standardized handling procedures make physical coupons an unreliable medium for managing repeated micro-transactions. Addressing this issue requires a shift from analog record-keeping to a digitized, tamper-proof system—one that TeaTech is specifically designed to provide.

1.2.3. Zero Auditability

A fundamental weakness of traditional paper-based coupon systems is their complete lack of auditability. Auditability refers to the ability to systematically trace, verify, and reconstruct the sequence of events involved in a transaction. In structured financial systems, audit logs serve as an essential layer of transparency, enabling stakeholders to identify errors, resolve disputes, and maintain trust. However, in the manual coupon exchange model used in many corporate vending environments, such mechanisms are entirely absent. This results in a transactional ecosystem where no party can reliably account for past interactions, making the system inherently fragile and prone to unresolved discrepancies.

The core reason for this absence of auditability lies in the unrecorded nature of coupon exchanges. When an employee hands over a physical coupon to the vendor, the exchange is immediate and ephemeral. There is no timestamp, no record of the quantity redeemed, and no documentation linking the redemption to a specific individual or event. Once the coupon changes hands, it becomes part of a loosely maintained stack that holds no meaningful metadata. This lack of traceability renders post-transaction verification impossible. In the event of disagreement—whether concerning the number of coupons remaining, discrepancies in daily totals, or questions regarding unauthorized deductions—neither users nor vendors have access to objective evidence to support their claims.

In most corporate systems, accountability relies on the ability to reconstruct past actions. However, in the paper coupon model, transaction invisibility is the norm. Unlike digital systems where every action can be logged, retrieved, and inspected, paper coupons provide no mechanism for retrospective analysis. Employees cannot review their redemption history, and vendors cannot conduct periodic audits to identify unusual patterns or possible discrepancies. As a result, when inconsistencies occur, both parties must rely on memory and informal communication, which are notoriously unreliable for accurate financial reconciliation.

The lack of auditability also creates challenges during large-scale reconciliation, such as at the end of the month when vendors and corporate administrators attempt to settle cumulative coupon usage. Since vendors depend on manually counted stacks of physical coupons, any loss, miscategorization, or miscount directly alters their reported totals. Without a digital ledger or identity-linked records, administrators are forced to accept the vendor's count as-is, even if it may contain unnoticed errors. This introduces systemic vulnerabilities that can distort financial reporting and cause avoidable financial discrepancies.

From an employee perspective, the absence of a verifiable transaction trail can be particularly frustrating. Employees often purchase coupon bundles in bulk, expecting them to last over several days or weeks. When their coupons deplete faster than anticipated, they have no means to determine whether the deficit resulted from accidental misuse, misplacement, or a vendor's counting error. This lack of transparency erodes user confidence and creates a perception of systemic unreliability, even when errors are unintentional. Over time, the psychological impact of not being able to verify one's own usage leads to reduced engagement with the system and reluctance to purchase coupons in advance.

Furthermore, zero auditability introduces vulnerabilities that make the system susceptible to potential misuse, even if such cases are rare. Without a verifiable chain of custody for coupons, scenarios such as double-counting, unrecorded deductions, or misuse of damaged coupons cannot be effectively prevented or detected. Although most vendors and employees operate in good faith, the system's design provides no safeguard against mistakes that mimic fraudulent activity. This creates an environment where trust must compensate for the absence of evidence—a situation that is inherently unstable in any transactional workflow.

In organizational contexts that emphasize compliance, accountability, and process integrity, the absence of auditable records is a significant drawback. Modern enterprises often require traceable data even for small-scale internal operations to ensure consistency and fairness. The paper coupon system stands in stark contrast to these expectations, functioning as an isolated manual process that cannot be integrated with corporate audit procedures. As organizations transition toward digital-first operational cultures, systems lacking auditability increasingly appear anachronistic and inefficient.

Ultimately, the zero-auditability problem highlights the fundamental need for a digitized transparency layer in corporate vending systems. A modern solution must provide chronological logs, user-specific histories, and vendor-side reconciliation tools that collectively eliminate ambiguity. TeaTech directly responds to this need by delivering a system where each transaction is recorded, timestamped, and verifiable, transforming an opaque process into a fully accountable workflow. Addressing auditability is not merely an enhancement—it is a prerequisite for building trust, reducing disputes, and aligning vending operations with contemporary corporate standards.

1.3. Proposed Solution: TeaTech Web Portal

1.3.1. Zero Payment Friction (USP)

A defining feature of TeaTech, and arguably its strongest unique selling proposition (USP), is its principle of **Zero Payment Friction**. This design philosophy emphasizes that the platform should digitize only the coupon validation process, while *intentionally avoiding* the integration of online payment gateways such as UPI, cards, or net banking. This strategic choice is not a limitation but a deliberate alignment with the operational realities of hyper-local vendors and the cultural norms that govern small-scale corporate vending ecosystems. By maintaining the simplicity of cash-based coupon purchases while digitizing redemption, TeaTech achieves a balance between modernization and practical usability—making it inherently more adoptable and less intrusive than conventional digital payment systems.

One of the primary motivations for adopting zero payment friction is the recognition that many corporate beverage vendors operate as micro-enterprises with minimal digital infrastructure. Unlike larger commercial vendors, these small-scale operators often do not possess the financial literacy, taxation compliance frameworks, or technical readiness to engage with full-scale digital payment systems. Integrating payment gateways would require vendors to handle settlement delays, transaction fees, bank reconciliation, and compliance with financial regulations—elements that complicate their routine business operations. By removing payment processing entirely from the TeaTech workflow, the platform ensures that vendors can continue accepting payments in their familiar manner while benefiting from digital efficiency during coupon redemption.

From the employee's perspective, zero payment friction removes barriers associated with digital transactions that might otherwise inhibit adoption. Employees typically purchase coupon bundles directly from the vendor using cash or any informal payment method already established between them. This flexibility allows the system to remain compatible with a wide range of user preferences while avoiding the overhead of integrating formal financial systems. When users are not compelled to set up digital wallets, link bank accounts, or navigate digital payment issues, the adoption curve becomes significantly smoother. This is especially beneficial in diverse workplaces where digital proficiency varies among employees.

At a structural level, zero payment friction also contributes to the **system's robustness and simplicity**. Digital payment systems introduce several challenges, including failed transactions, double charges, integration breakdowns, and dependency on external APIs. Such failure points would increase system complexity and maintenance overhead while diluting the platform's core purpose. By excluding payment processing entirely, TeaTech reduces the number of potential vulnerabilities and focuses on delivering a seamless, reliable, and low-maintenance solution centered exclusively on coupon management. This strategic minimization strengthens both performance and reliability.

Moreover, this USP directly addresses the trust-related problems inherent in traditional coupon systems. Employees and vendors often hesitate to shift to digital platforms due to fears that digital payments may complicate an otherwise straightforward exchange. By limiting digitization to the validation stage, TeaTech avoids interfering with established financial habits. Instead, it enhances trust by providing a

secure, transparent method for tracking coupon usage while preserving the autonomy of existing vendor–employee financial interactions. In essence, it modernizes what needs modernization without altering what already works well.

Zero payment friction also supports **faster deployment and broader scalability**. Since regulatory compliance, payment gateway licensing, and financial data security requirements are not involved, the system can be deployed in virtually any workplace without additional legal or infrastructural constraints. The absence of payment integration not only reduces development effort and cost but also enables organizations to adopt the platform quickly and with minimal resistance from stakeholders. Vendors who might otherwise be hesitant to adopt a fully digital commerce system are more receptive to a solution that augments their workflow without rewriting it.

Additionally, the decision to avoid payment integration reflects a deep understanding of the real-world constraints observed during the initial pilot testing phase. Many vendors expressed reluctance to migrate to systems that tied their earnings to digital settlement cycles, and employees preferred retaining flexibility in how they purchased coupon bundles. These human-centered insights strongly supported the decision to pursue zero payment friction as a core principle rather than as an optional feature. The outcome is a system that aligns closely with the lived experiences, comfort levels, and operational norms of all stakeholders involved.

In summary, zero payment friction is not merely a technical omission but a **strategic innovation** that strengthens TeaTech's viability in real-world scenarios. By preserving the simplicity of cash-based vendor operations and eliminating barriers associated with online payments, TeaTech ensures high adoption rates, reduces operational complexity, and delivers a solution finely tuned to the needs of hyper-local vending environments. This USP distinguishes TeaTech from conventional digital vending platforms and establishes it as a practical, accessible, and trust-enhancing system for corporate beverage transactions.

1.3.2. Two-Step Confirmation

A central component of TeaTech's solution framework is its **Two-Step Confirmation** mechanism, designed to eliminate unauthorized deductions, miscommunication errors, and ambiguity during coupon redemption. This mechanism introduces a shared verification step between the vendor and the employee, ensuring that both parties explicitly agree to each transaction before it is finalized. In doing so, TeaTech addresses one of the most critical shortcomings of traditional paper-based systems: the absence of real-time validation and user involvement in the deduction process.

In conventional coupon exchanges, the vendor independently marks or collects a coupon from the employee, often amidst crowded and fast-paced environments. This single-point control introduces multiple risks, such as accidental deductions, miscounted coupons, or misunderstandings during verbal confirmation. Employees may forget whether they handed over a coupon, vendors may process deductions too quickly, or environmental noise may distort communication. The two-step confirmation process counteracts these risks by shifting the system to a **dual-interaction model**, where the vendor initiates the transaction but the employee authorizes it.

The mechanism begins when the vendor inputs a request to deduct a specific number of coupons from a user's digital balance. Instead of immediately reducing the balance, the system generates a pending request visible to the employee in real time. Only after the employee reviews and approves the request does the actual deduction occur. This structured delay creates a deliberate checkpoint that prevents unintentional or unauthorized transactions. In effect, the system elevates user agency by ensuring that employees actively validate every redemption, thereby fostering a sense of transparency and control.

One of the key advantages of the two-step confirmation model is its ability to **eliminate reliance on verbal cues**, which are often the source of inconsistencies in high-traffic vending environments. During busy hours or in noisy surroundings, quick interactions like nods, gestures, or fragmented dialogue can easily lead to misunderstandings. The confirmation request displayed on the user interface replaces these informal cues with a clear, unambiguous digital prompt. This not only reduces communication errors but also creates a standardized process that remains consistent regardless of contextual factors such as volume of customers or vendor workload.

The implementation of two-step confirmation also aligns with best practices in secure transactional design, where critical operations require acknowledgment from multiple parties. By extending this concept to micro-transactions, TeaTech incorporates principles such as **redundancy, explicit consent, and verification** into everyday beverage purchase workflows. These principles are commonly observed in banking, authentication systems, and enterprise software, but have traditionally not been applied to small-scale vending operations. Introducing them in this context significantly enhances trust and accountability while maintaining the system's simplicity.

From a practical standpoint, the two-step mechanism also serves as a built-in **error detection layer**. Users may occasionally request a different quantity than what the vendor perceives, or vendors may accidentally input incorrect values. Without a confirmation step, such discrepancies could lead to disputes or inaccurate deductions. With TeaTech's workflow, employees can immediately detect inconsistencies and either approve or decline the request. Declined requests help vendors correct mistakes on the spot, thereby improving operational accuracy and ensuring that the transaction only proceeds with mutual agreement.

Furthermore, the confirmation process creates a **digitally logged trail of activity**, enabling complete transparency. Each confirmation event is stored with a timestamp, user identity, vendor identity, and deduction details. These structured logs not only assist in resolving disputes but also reinforce system integrity by providing an objective reference for audits and historical reviews. In environments where accountability has traditionally been difficult to maintain, this explicit recordkeeping becomes an essential component of operational reliability.

Pilot testing in a real-world corporate internship environment demonstrated strong user acceptance of the two-step confirmation model. Employees reported a heightened sense of confidence knowing that deductions required their explicit approval. Vendors appreciated that the system helped prevent accidental over-deductions during busy hours, reducing friction and misunderstandings. These observations validated the necessity of this mechanism and highlighted its practicality in actual vending scenarios.

In summary, the two-step confirmation system is a transformative element of TeaTech's operational workflow. By requiring dual consent, replacing informal verbal cues with digital prompts, and establishing a verifiable transaction trail, it directly addresses the core weaknesses of existing paper coupon processes. This mechanism not only enhances security and accuracy but also strengthens user-vendor trust—making it a foundational pillar of the TeaTech solution.

1.4. Project Scope and Limitations

1.4.1. Scope

The scope of TeaTech defines the functional boundaries, intended objectives, and operational contexts within which the system is designed to perform. Establishing a clear scope ensures that the project remains focused on addressing the specific challenges associated with corporate tea and coffee coupon management while avoiding unnecessary complexity. TeaTech is deliberately crafted as a lightweight, accessible, and highly practical solution that modernizes the coupon redemption layer without altering the financial workflows of hyper-local vendors. The system's scope reflects its targeted purpose: to replace fragile, error-prone paper coupons with a secure, transparent, and user-centric digital alternative.

At its core, TeaTech provides employees with a streamlined interface to maintain their digital coupon balance, monitor redemption activity, and approve vendor-initiated deductions. The platform empowers users by giving them real-time visibility and control over their coupon usage, features that were entirely absent in traditional paper-based systems. Employees can track their remaining coupons, review past transactions, and interact with the system in both English and Hindi, ensuring accessibility across diverse user groups. This multilingual adaptability broadens the system's reach and enhances usability for users who may not be proficient in English.

On the vendor side, TeaTech supports a simplified operational interface tailored to the fast-paced nature of beverage counters. Vendors can initiate deduction requests, view pending confirmations, and track completed transactions without navigating complex features. The system is optimized for minimal cognitive load, allowing vendors to focus primarily on service delivery rather than administrative tasks. Importantly, TeaTech supports multiple vendors within a corporate environment, ensuring scalability in settings where several beverage counters or stalls operate simultaneously.

A central element of the project's scope is the implementation of TeaTech's **two-step confirmation workflow**, which governs all coupon-related interactions. This workflow ensures that every deduction undergoes explicit consent from both vendor and employee, thereby addressing common issues such as miscommunication, accidental deductions, and counting errors. The scope of the confirmation mechanism includes real-time synchronization, request validation, and atomic transactions through Firebase Firestore to maintain data consistency. These technical capabilities form a foundational part of TeaTech's operational design.

The scope further extends to providing a **tamper-proof audit trail** for all transactions. By leveraging a NoSQL real-time database, TeaTech maintains structured logs of coupon activities, including timestamps,

user identities, vendor identities, and transaction outcomes. This feature supports transparency and accountability, enabling users and vendors to reference past interactions whenever necessary. Such comprehensive logging also facilitates future organizational audits or performance analysis by administrative authorities, if integrated into workplace management systems.

While TeaTech digitizes the coupon redemption process, its scope intentionally excludes payment processing. Users continue to purchase coupon bundles directly from the vendor using their preferred offline or informal method. This decision reflects the platform's emphasis on simplicity, immediate usability, and low technological barriers for small-scale vendors. By avoiding online payments, TeaTech preserves the financial autonomy of the vendor while ensuring that the system remains cost-effective and easy to adopt.

From a deployment perspective, the scope includes operation across standard web browsers and mobile-responsive interfaces. TeaTech is designed to function without specialized hardware, making it compatible with a wide range of devices including smartphones, tablets, and low-cost vendor terminals. This flexibility ensures that organizations can deploy the system rapidly without significant infrastructural investment. Moreover, the Firestore backend supports real-time data updates, enabling seamless functionality even in environments with modest network connectivity.

The scope also encompasses the platform's adaptability for future enhancements. Although the initial version is centered on coupon validation and transaction transparency, the system architecture allows for the integration of additional features such as location-based vendor sorting, vendor analytics dashboards, and corporate-level administrative tools. These capabilities lie outside the immediate implementation but fall within the system's extended conceptual scope, ensuring that TeaTech remains scalable and upgrade-friendly.

In summary, the scope of TeaTech includes digital coupon balance management, vendor-initiated deductions, two-step confirmations, bilingual interfaces, audit trails, and scalable multi-vendor support—while deliberately excluding digital payments to preserve operational simplicity. This focused yet flexible scope ensures that TeaTech meaningfully addresses the core challenges of paper-based coupon systems while remaining lightweight, practical, and aligned with the real-world needs of corporate vending environments.

1.4.2. Limitations

While TeaTech provides a structured and effective solution for digitizing coupon redemption in corporate vending environments, it is essential to acknowledge the inherent limitations that define its operational boundaries. Identifying these limitations not only clarifies the constraints of the current system but also helps contextualize future improvements. These limitations arise from conscious design decisions, practical constraints of the deployment environment, and the scope of technology intentionally selected to keep the system lightweight and easily adoptable.

A primary limitation of TeaTech is the **absence of integrated digital payment functionality**. Although this is deliberate—aligned with the principle of zero payment friction—it also means that users must continue purchasing coupon bundles directly from vendors using offline payment methods. For

organizations seeking a fully automated end-to-end financial system, TeaTech in its current form does not handle monetary settlements, payment verification, or transaction reconciliation between employee accounts and vendor revenue. The system digitizes coupon validation, not coupon purchasing, which may be viewed as a limitation in contexts where end-to-end digitization is expected.

Another inherent limitation lies in TeaTech's reliance on **stable internet connectivity** for real-time synchronization. Since the platform depends on Firebase Firestore to update balances, manage confirmation requests, and store transaction logs, intermittent connectivity may temporarily impair its functionality. In environments with inconsistent network availability, users may experience delays in receiving confirmation prompts, and vendors may face interruptions when sending deduction requests. Although Firestore offers limited offline persistence, the two-step confirmation workflow fundamentally depends on real-time communication between user and vendor devices.

The system's dependence on **device accessibility** also introduces constraints. Vendors and employees must have access to a functional smartphone, tablet, or computer to interact with the platform. While this requirement is reasonable in most modern corporate environments, it may pose challenges in lower-resource settings or for users unfamiliar with digital interfaces. Moreover, if a vendor's device malfunctions, lacks sufficient battery, or experiences browser-related issues, TeaTech's functionality may be temporarily disrupted until the device becomes operational again.

A further limitation involves the platform's **restricted applicability outside corporate or semi-structured environments**. TeaTech is optimized for workplaces where employees and vendors have established interactions and where coupon-based systems already exist. In open public marketplaces, street-side stalls, or non-institutional settings, user–vendor relationships and transaction volumes may not align with the structured workflow TeaTech requires. As a result, the system's effectiveness is closely tied to environments that support predictable user flows and controlled operational dynamics.

Additionally, TeaTech currently lacks **role-based administrative tools** for higher-level oversight. Corporate HR departments, facility managers, or kiosk supervisors do not have a centralized dashboard in the initial version to monitor vendor performance, coupon distribution trends, or user engagement metrics. While such features fall within the future scope of the project, their absence in the current implementation limits the system's immediate utility for enterprise-level analytics or administration.

The platform also faces limitations related to **error handling in edge cases**. For instance, if a user accidentally approves a deduction or if a vendor mistakenly initiates an incorrect request, the system does not currently include an automated reversal mechanism. Users must rely on vendor discretion or manual adjustment for reconciliation. Although this preserves workflow simplicity, it may lead to occasional inconvenience in scenarios where rapid rectification is needed.

Another constraint arises from TeaTech's **single-transaction focus**, which currently supports only coupon deductions and not additional workflow variations such as coupon transfers, bundle refunds, or multi-item redemption logic. While these features are not essential for the core problem TeaTech solves, their absence restricts the system's versatility in handling more complex vending operations.

Finally, the system has a limited **visual reporting and analytics interface**. Users receive detailed transactional logs, but advanced features such as summarized spending patterns, heat maps of vendor activity, or predictive consumption trends are not part of the present implementation. This restricts the platform's utility to operational transparency rather than strategic insight.

In summary, TeaTech's limitations stem from consciously scoped design choices, technological dependencies, and the operational realities it aims to address. While these constraints do not undermine the system's effectiveness in solving the core problems of paper-based coupon management, they do outline important areas for refinement and expansion in subsequent iterations. Recognizing these limitations provides a realistic perspective on TeaTech's current capabilities and sets the stage for future development.

1.5. Organization of the Report

This project report is structured into seven comprehensive chapters, each addressing a specific dimension of the TeaTech system—from conceptual foundations and literature background to system design, implementation, and future enhancements. The organization follows a logical progression that mirrors the development lifecycle of the project, ensuring clarity, coherence, and ease of understanding for readers. By systematically presenting the motivations, technical foundations, methodologies, and outcomes, the report provides a holistic perspective on how TeaTech was conceptualized, developed, tested, and evaluated.

Chapter 1 introduces the project by outlining the motivation behind TeaTech, the challenges inherent in traditional paper-based coupon systems, and the rationale for adopting a digital solution. It also details the project's scope and limitations, establishing the contextual foundation necessary for understanding the system's objectives and constraints. This chapter concludes with an overview of how the remaining sections of the report are organized.

Chapter 2 presents a detailed **literature survey**, situating TeaTech within the broader landscape of coupon management systems, micro-payment platforms, and digital vending technologies. It reviews relevant studies, existing commercial systems, and traditional mechanisms to highlight the evolution of vending solutions. Through gap analysis, the chapter identifies limitations in current offerings and demonstrates how TeaTech contributes uniquely to this domain by addressing transparency, accountability, and ease of adoption.

Chapter 3 focuses on **system analysis and requirements engineering**, outlining both functional and non-functional requirements essential for TeaTech's operation. It delves into the core features such as user authentication, real-time balance management, vendor linking, two-step confirmation workflows, and bilingual support. The non-functional requirements further define expectations related to performance, security, usability, and scalability. Together, these requirements form the blueprint for the system's design.

Chapter 4 describes the system's **design architecture**, presenting the structural and logical components that define TeaTech's operation. This includes a three-tier architecture covering the frontend, backend logic, and Firestore database. The chapter also details the data model, core entities, and workflow

diagrams, including the two-step confirmation process. These design elements illustrate how the system translates conceptual requirements into an actionable technical framework.

Chapter 5 elaborates on the **implementation details**, providing insight into the technology stack, development methodologies, and key logic components. It includes an explanation of the coupon deduction function, pseudocode representations, and the atomic transaction mechanisms used to maintain data consistency. The chapter also discusses the bilingual interface implementation strategy based on JSON localization files, reflecting TeaTech's focus on usability and inclusiveness.

Chapter 6 examines the **testing methodologies and results**, covering unit testing, integration testing, and user acceptance testing conducted during the project's deployment in a real-world corporate internship environment. This chapter presents critical test scenarios—particularly those involving the two-step confirmation workflow—and evaluates the system's performance based on quantitative metrics and user feedback. The results confirm TeaTech's effectiveness in reducing errors and improving transactional transparency.

Chapter 7 provides the **conclusion and outlines the future scope** of the project. It summarizes the key findings, highlights the system's contributions to solving real-world operational challenges, and discusses potential enhancements such as location-based vendor sorting, progressive web app (PWA) capabilities, analytics dashboards, and extended API integrations. These improvements signify the system's scalability and readiness for broader application.

Overall, the organization of this report ensures that readers can follow the progression from identifying the core problem to implementing and validating a targeted technological solution. Each chapter builds upon the previous one, presenting a cohesive narrative that captures both the practical necessity and technical innovation embodied in TeaTech.

CHAPTER 2: LITERATURE SURVEY

2.1. Introduction to Digital Vending Systems

Digital vending systems have emerged as a transformative development in transactional technology, reshaping how goods and services are accessed, purchased, and consumed across various sectors. Unlike traditional vending mechanisms that rely heavily on manual processes and physical tokens, digital vending systems leverage modern software platforms, real-time communication channels, and cloud-based infrastructures to automate and streamline micro-transactions. These systems are particularly valuable in environments characterized by high transaction volumes, rapid service turnover, and the need for accurate, traceable, and user-friendly operational workflows. As organizations increasingly embrace digital solutions to enhance efficiency and transparency, digital vending systems have become a cornerstone of contemporary service delivery models.

At their core, digital vending systems integrate user authentication, inventory management, transaction processing, and audit trails into a cohesive digital ecosystem. Users interact with these systems through mobile applications, web portals, RFID tags, or smart cards to initiate or complete transactions. The system's backend processes these interactions by validating credentials, deducting credits, updating inventory, and ensuring that every operation adheres to predefined business rules. Modern implementations often utilize cloud technologies to maintain real-time synchronization between user devices and service terminals, enabling seamless interactions regardless of scale.

A defining characteristic of digital vending systems is their emphasis on **automation and accuracy**. Traditional vending workflows—such as paper coupon redemption, manual cash collection, or handwritten logs—are susceptible to human error, miscommunication, and fraud. Digital systems minimize these risks by implementing structured, rule-driven workflows that eliminate ambiguity and maintain precise records of every transaction. This automated reliability is one of the primary reasons digital vending is increasingly adopted in both corporate and public environments.

Digital vending systems also play a crucial role in enhancing **user experience and convenience**. Users can access their transaction history, check balances, receive notifications, and make real-time decisions through intuitive digital interfaces. These systems cater to a range of user preferences and support multilingual environments, making them accessible to diverse populations. In many implementations, biometric or two-factor authentication mechanisms further enhance security and accuracy of the vending process.

From an operational standpoint, digital vending systems provide vendors and service providers with tools to manage their workflows more efficiently. Vendors can monitor sales volumes, peak operational hours, inventory levels, and customer behavior through integrated dashboards or analytics tools. These insights allow them to optimize resource allocation, forecast demand more accurately, and improve service delivery. Additionally, digital systems reduce the administrative burden of manual recordkeeping, enabling vendors to focus on service provision rather than operational overhead.

Historically, digital vending systems have evolved alongside advancements in mobile computing, cloud infrastructure, and decentralized identity management. Early digital vending solutions primarily focused on cashless payments through magnetic cards or RFID tags. As mobile devices became ubiquitous, these systems transitioned to app-based interfaces with enhanced capabilities such as QR scanning, digital wallets, remote authentication, and automated reconciliation. Today's systems often integrate artificial intelligence, IoT sensors, and edge computing to deliver intelligent and adaptive vending experiences.

In the corporate landscape, digital vending systems have gained particular relevance. Large office campuses, technology parks, and industrial complexes often manage thousands of daily micro-transactions involving food, beverages, printing services, transport tokens, and more. Digital vending systems help organizations ensure operational transparency, prevent financial disputes, and provide employees with reliable access to essential services. The integration of audit logs and real-time data synchronization further aligns digital vending systems with corporate governance and compliance standards.

The relevance of digital vending systems to TeaTech becomes evident when considering the limitations of traditional paper-based coupon systems. Although paper coupons are simple, they lack auditability, durability, scalability, and error mitigation mechanisms. Digital vending principles—such as real-time updates, user-involved validations, structured workflows, and transaction logging—directly address these issues. TeaTech applies the foundational concepts of modern digital vending within a hyper-local corporate context, offering a minimalistic yet effective solution tailored to the needs of beverage vendors and employees.

In summary, digital vending systems represent a convergence of automation, cloud computing, user-centric design, and secure transaction workflows. Their evolution reflects a broader shift toward digitization across service-oriented industries. By drawing from these established concepts and adapting them to the micro-interaction space of corporate beverage counters, TeaTech demonstrates how digital vending principles can be successfully applied to small-scale but operationally significant workflows.

2.2. Review of Related Works

2.2.1. Loyalty and Coupon Management Systems

Loyalty and coupon management systems represent one of the earliest forms of structured customer engagement technologies. These systems enable organizations to incentivize user behavior through discounts, reward points, digital credits, or redeemable tokens. Over the last two decades, they have evolved from manual punch-card mechanisms to sophisticated, data-driven platforms integrated across mobile applications, cloud services, and customer analytics engines. Although these systems differ in scale and purpose from corporate beverage coupon workflows, their underlying principles—such as transaction validation, credit redemption, and user engagement—serve as valuable reference points for the development of TeaTech.

Traditional loyalty systems began with **physical stamp cards or paper coupons**, where customers collected proof of purchase to redeem rewards. While simple to deploy, such systems suffered from issues similar to those seen in corporate paper coupon workflows: loss of physical tokens, lack of audit trails, vulnerability to fraud, and an inability to enforce usage rules. These limitations accelerated the shift toward digital coupon systems, where users maintain accounts tied to phone numbers, customer IDs, or loyalty cards, eliminating the dependency on physical artifacts. The digital transition provided organizations with improved accuracy, better tracking, and enhanced customer retention strategies.

Modern digital loyalty systems operate through **centralized or cloud-based platforms** that manage user points, coupon issuance, redemption tracking, and rule enforcement. These platforms often integrate with point-of-sale terminals, QR-code scanners, or NFC-based devices to streamline redemption. The core advantage lies in the creation of **verifiable, timestamped transaction logs**, which ensure transparency for users and accountability for service providers. Such audit capabilities closely parallel the foundational objectives of TeaTech, where the need for traceability and error-free redemption is paramount.

A notable feature of contemporary loyalty systems is their emphasis on **user-centric design**, ensuring that customers can view their balances, rewards, and transaction histories at any time. This visibility helps users make informed decisions and builds trust in the system's reliability. TeaTech adopts a similar philosophy by providing employees with real-time access to their digital coupon balance and comprehensive transaction history. These capabilities directly address the transparency gaps inherent in paper coupon systems, demonstrating how principles from large-scale loyalty frameworks can be effectively adapted to micro-vending environments.

Digital loyalty systems also frequently incorporate **multi-language support** to ensure accessibility across diverse user populations. Whether deployed in retail, transportation, hospitality, or corporate settings, multilingual interfaces enhance usability and broaden adoption. TeaTech draws from this approach by integrating bilingual support (Hindi and English), recognizing the linguistic diversity present in many Indian workplaces and ensuring that both employees and local vendors can interact with the platform comfortably.

Another parallel can be drawn from the **security models** utilized in loyalty platforms. Modern systems employ authentication protocols, unique user identifiers, and anti-fraud mechanisms to prevent unauthorized use of digital credits. Some systems utilize OTP-based verification or require user approval before redeeming rewards. TeaTech's two-step confirmation mechanism reflects a similar security approach, ensuring that no deduction occurs without explicit employee consent. This mechanism represents a refined adaptation of best practices observed in broader digital redemption ecosystems.

However, despite these similarities, many large-scale loyalty and coupon management systems integrate advanced features such as dynamic pricing, personalized offers, machine learning-based recommendations, and marketing automation. These functions serve commercial objectives but exceed the functional requirements of a simple beverage coupon workflow. TeaTech consciously avoids such complexity to maintain minimalism and user-friendliness, making it more suited to small-scale vending operations where speed and simplicity are essential.

Moreover, loyalty programs typically include **payment integration**, allowing customers to earn or redeem points based on monetary transactions. TeaTech diverges from this model by adhering to its principle of zero payment friction. While payment-enabled loyalty platforms require transaction fees, settlement structures, and financial compliance, TeaTech simplifies adoption by focusing exclusively on coupon validation without altering financial workflows. This distinction highlights TeaTech's context-specific design, emphasizing operational practicality over commercial sophistication.

In summary, loyalty and coupon management systems offer valuable insights into the design principles that underpin effective digital redemption frameworks: user visibility, transaction logging, multilingual usability, and secure verification. TeaTech incorporates these principles while adapting them to the constraints and expectations of hyper-local corporate vending. The project diverges where necessary—most notably in its exclusion of digital payments—to ensure relevance, simplicity, and rapid adoption in environments where traditional coupon systems have long prevailed.

2.2.2. Micro-Payment and UPI Systems

Micro-payment systems and Unified Payments Interface (UPI) platforms have revolutionized the digital transaction landscape, particularly in countries like India where cashless payments have seen unprecedented adoption. These systems are engineered to support low-value, high-frequency transactions with minimal latency, making them suitable for a wide range of day-to-day financial activities. Although their primary focus differs from TeaTech's operational scope, micro-payment and UPI-based systems offer a useful comparative framework for understanding the strengths and trade-offs of alternative digital transaction models in contexts similar to corporate beverage vending.

Micro-payment systems are designed to facilitate transactions that are too small to be efficiently processed through traditional payment channels. Historically, such transactions faced obstacles due to high transaction fees and long settlement cycles imposed by banking systems. The emergence of digital wallets, prepaid accounts, and mobile payment applications resolved many of these issues by enabling instantaneous transfers with negligible overhead. These platforms rely on centralized transaction servers, encryption protocols, and user authentication mechanisms to maintain security and ensure reliability across millions of concurrent users.

In India, **UPI has become the de facto standard for micro-payments**, enabling seamless peer-to-peer (P2P) and peer-to-merchant (P2M) transfers. UPI's architecture is built on interoperability, meaning users can transact across different banks and payment applications through a unified protocol. Features such as real-time settlement, QR-code scanning, virtual payment addresses (VPAs), and instant payment notifications make UPI exceptionally convenient for everyday transactions, including small purchases such as snacks, beverages, or cafeteria items within corporate environments.

However, despite its efficiency, UPI introduces a layer of **financial formality and regulatory compliance** that is often incompatible with hyper-local vendors who prefer cash-based operations. Digital transactions require vendors to maintain linked bank accounts, manage reconciliation reports, pay potential transaction fees, and satisfy verification requirements imposed by payment gateways. For many

small beverage stall operators, these obligations are perceived as barriers rather than conveniences. As a result, adoption remains uneven among micro-vendors even in digitally advanced regions.

This contrast is significant when viewed through the lens of TeaTech's design objectives. Although UPI and other micro-payment systems achieve remarkable accuracy and visibility, they impose infrastructural, technical, and financial expectations that exceed what is necessary for a simple coupon-based beverage system. TeaTech's founders, during the initial problem observation phase, noted that many vendors explicitly expressed discomfort with digital payments due to settlement delays, possible bank disputes, or the perceived intrusion of financial tracking into their daily cash flow. These insights helped shape TeaTech's commitment to **zero payment friction**, distinguishing it fundamentally from micro-payment platforms.

Another challenge in applying UPI to coupon workflows lies in the **transactional overhead for both parties**. For an employee purchasing a ₹10 tea or coffee multiple times a day, initiating a UPI payment each time introduces unnecessary cognitive and operational friction. Users must unlock their mobile device, open a payment app, scan a QR code or enter details, confirm the transfer, and wait for acknowledgment before receiving the beverage. In high-volume corporate environments, this lengthens queues and disrupts the intended quick-service nature of vending counters. TeaTech, by contrast, separates financial exchange from redemption, enabling employees to pre-purchase coupons in bulk and redeem them effortlessly with only a confirmation tap during service.

UPI and digital wallets also operate under a **centralized financial ledger** maintained by banks and regulated entities. While this assures compliance and financial integrity, it constrains customization. Vendors cannot modify transaction workflows to suit niche use cases such as multi-deduction coupons, loyalty redemptions, or bundled offers without integrating additional systems or developing proprietary layers. TeaTech, built specifically for beverage coupon workflows, has the flexibility to tailor its logic, UI design, and workflow strategies to the exact needs of corporate vending counters.

Despite these differences, micro-payment systems provide several conceptual parallels that strengthen TeaTech's design insight. The emphasis on **real-time throughput**, **transaction security**, **user authentication**, and **mobile-first interface design** all serve as foundational inspirations. TeaTech replicates these strengths in a limited but focused manner—by ensuring quick confirmation flows, data integrity through Firestore transactions, and mobile responsiveness without overburdening the system with full-fledged financial processing capabilities.

Furthermore, micro-payment systems highlight the importance of **digital inclusivity**, addressing diverse user types ranging from tech-savvy employees to individuals with limited digital literacy. This reinforces TeaTech's own commitment to bilingual interfaces, intuitive layouts, and simplified workflows. While UPI applications contain multiple layers of functionality, TeaTech distills only what is necessary for a coupon redemption workflow, thereby maintaining high accessibility.

In summary, while micro-payment and UPI systems represent powerful and robust frameworks for digital financial transactions, their complexity, infrastructural expectations, and regulatory overhead make them misaligned with the needs of hyper-local beverage vendors. TeaTech instead adopts the most relevant principles—real-time digital updates, secure validation, and user-centric interfaces—while consciously

omitting payment integration to ensure rapid adoption, simplicity, and workflow compatibility. By understanding the strengths and constraints of micro-payment systems, TeaTech strategically positions itself as a minimalistic yet effective alternative tailored for small-scale corporate vending environments.

2.2.3. Traditional Paper-Based Coupon Systems

Traditional paper-based coupon systems have long been used across corporate environments, educational institutions, industrial campuses, and small retail setups as a simple and low-cost mechanism for managing micro-transactions. These systems typically rely on printed coupon booklets or loose paper tokens that employees purchase in advance and redeem at vendor stalls in exchange for beverages or snacks. While this model is valued for its simplicity and independence from digital infrastructure, it exhibits deep structural weaknesses that become increasingly evident in environments with high transaction volumes and rapid service cycles. Understanding these limitations provides essential context for TeaTech's development and illustrates the necessity of transitioning toward a digital alternative.

One of the most notable shortcomings of paper-based coupon systems is their **inherent susceptibility to human error**. Since coupons are manually issued, handled, and collected, discrepancies often arise due to incorrect counts, oversight, or hurried exchanges during peak hours. Vendors may inadvertently collect fewer or more coupons than intended, and employees may struggle to recall their remaining balance accurately. As transaction volumes increase, these inconsistencies accumulate, making dispute resolution extremely difficult. The lack of standardized procedures for coupon handling further intensifies the frequency of such errors.

In addition to human error, paper coupons suffer from pronounced **physical fragility**. Coupons are prone to tearing, folding, moisture damage, and loss. Employees frequently carry their coupon booklets in bags, pockets, or notebooks, exposing them to conditions that can easily destroy or displace them. Vendors often gather large stacks of redeemed coupons in small containers or loosely organized piles, where even minor disturbances—such as wind, spillage, or clutter—can lead to loss or mixing. These physical weaknesses introduce continuous uncertainty into the system, often forcing one party to bear the consequences of accidental loss without any means of verification.

A fundamental structural limitation of paper-based coupon systems is their **zero auditability**. Once a coupon is redeemed, it leaves no digital or traceable record linking the transaction to a specific employee, vendor, or timestamp. This absence of verifiable transaction history undermines transparency and trust. When discrepancies arise, both employees and vendors must rely solely on memory or informal communication, neither of which provides a reliable basis for reconciliation. Large organizations attempting end-of-month settlements face particular difficulty, as the vendor's reported coupon count becomes the only available reference—despite its vulnerability to error.

Traditional coupon systems also impose **administrative inefficiencies** on vendors. Throughout the day, vendors must collect, store, and sometimes sort large numbers of physical coupons, often while simultaneously serving customers. This process interrupts workflow, slows service during busy periods, and adds unnecessary cognitive load. At the end of each day or billing cycle, vendors must manually count collected coupons to determine revenue earned. Manual counting is time-consuming and

error-prone, especially when vendors manage hundreds of transactions daily. Without standardized logs or structured recordkeeping, mistakes can easily pass undetected.

From an organizational perspective, paper-based coupon systems create **operational friction** that affects both productivity and employee satisfaction. Employees who misplace coupons must purchase new booklets, resulting in financial loss and frustration. Vendors may experience conflicts when users contest discrepancies in coupon counts. These frequent micro-disputes, though individually small, accumulate into systemic inefficiencies that disrupt workplace harmony. In large corporate settings, where beverage counters experience high footfall, the cumulative impact becomes particularly noticeable.

Security limitations are another major concern in traditional coupon systems. Paper coupons can be **duplicated, forged, or reused** if not carefully monitored. Even if such misuse is rare, the system's design offers no safeguards against it. Because vendors have no mechanism to verify the authenticity or issuance of a coupon beyond visual inspection, counterfeit coupons can potentially enter circulation unnoticed. Organizations relying on paper coupons have no means of preventing or tracing such occurrences.

Moreover, traditional systems lack **scalability**. As organizations grow in size, the volume of coupon usage expands proportionally, placing increasing strain on vendors and employees. Manual counting becomes impractical, communication errors escalate, and storage becomes cumbersome. What may work adequately for a small office quickly becomes unsustainable for large IT parks or multinational corporate campuses where thousands of daily beverage transactions occur.

Finally, paper-based systems offer no **data-driven insights**. Unlike digital systems that can generate analytics on consumption trends, peak hours, vendor performance, or user expenditure patterns, paper coupons provide no structured data for analysis. Organizations looking to optimize resource allocation or evaluate vendor efficiency have no quantitative information to support decision-making. This lack of visibility contrasts sharply with modern organizational expectations centered on accountability, transparency, and measurable performance metrics.

In summary, traditional paper-based coupon systems, though simple and culturally familiar, suffer from structural fragility, high error rates, zero auditability, administrative inefficiencies, security vulnerabilities, and lack of scalability. These limitations make them increasingly unsuited for modern corporate environments, where reliability and transparency are essential. TeaTech emerges as a direct response to these shortcomings, offering a digital solution that preserves the simplicity of coupon-based workflows while eliminating the systemic flaws that hinder operational effectiveness.

2.3. Gap Analysis and Project Contribution

Gap analysis provides a structured method for identifying the limitations of existing systems and evaluating how a new solution addresses these deficiencies. Within the context of corporate beverage vending, traditional paper coupon systems, loyalty platforms, and micro-payment solutions each offer certain advantages but fall short in key operational areas. By examining these shortcomings, it becomes clear why a specialized system like TeaTech is necessary, and how its design directly contributes to resolving the unresolved challenges of coupon-based workflows.

The first significant gap emerges from the **lack of transparency and auditability** in traditional paper coupon systems. As discussed in the literature, physical coupons cannot store transaction metadata, making it impossible to reconstruct historical usage or resolve disputes effectively. Neither employees nor vendors possess verifiable records to confirm how many coupons were issued, redeemed, or lost. Existing loyalty systems address this through digital logs but are designed primarily for commercial retail interactions, not for micro-vending scenarios where fast-paced, low-value transactions occur continuously. TeaTech fills this gap by providing a complete, tamper-proof audit trail for every coupon deduction, enabling both parties to verify transaction histories with confidence.

Another major gap involves the **high error rate and friction associated with manual workflows**. Traditional coupons rely on human judgment, memory, and physical handling—all of which are error-prone, especially in high-traffic environments. Loyalty programs do improve accuracy through digital systems, but they typically require complex integrations, customer databases, and payment linkages that are inappropriate for small-scale vendors. Micro-payment systems such as UPI eliminate physical handling but introduce excessive interaction steps and impose financial compliance burdens. TeaTech bridges this gap by offering a digital redemption process that is both lightweight and operationally aligned with how vendors and employees naturally interact.

A third gap concerns the **adoption barrier** created by payment integration in existing digital solutions. UPI, digital wallets, and loyalty programs require users and vendors to engage with banking systems, manage refunds or settlements, and navigate technological requirements that extend beyond the needs of a simple beverage transaction. Many local vendors prefer cash-based workflows due to their immediacy, familiarity, and independence from regulatory oversight. TeaTech's principle of zero payment friction directly addresses this gap by allowing users to buy coupon bundles offline while digitizing only the redemption layer. This ensures rapid adoption with minimal behavioral change.

A further gap lies in the **limited user involvement** in conventional coupon systems. Paper workflows do not allow employees to validate deductions before they occur, resulting in disputes and uncertainty. Even digital systems that automate redemption often focus on point-of-sale scanning or automated deduction without requiring user confirmation. TeaTech uniquely contributes a **two-step confirmation mechanism**, ensuring that deductions occur only after explicit employee approval. This innovation addresses a critical gap in micro-vending environments, where miscommunication and crowd dynamics frequently lead to accidental or incorrect deductions.

Another identified gap pertains to **language accessibility**. Many existing digital vending or loyalty systems are designed for urban retail markets and assume English-language proficiency. In contrast, corporate environments—particularly in India—include diverse linguistic groups, including vendors who may prefer Hindi or other regional languages. TeaTech fills this gap by offering **bilingual support**, enabling seamless usability for both employees and vendors. This reduces the learning curve, improves adoption rates, and ensures equitable access across user groups.

From a technological perspective, existing solutions often lack **modularity and scalability** tailored for micro-vending needs. Loyalty platforms are optimized for large businesses, while UPI systems serve broad financial markets. Neither offers customizable workflows suitable for beverage counters within corporate campuses. TeaTech's architecture—built on Firebase Firestore—provides real-time

synchronization, lightweight deployment, and automated conflict resolution through atomic transactions, offering a scalable digital alternative without heavy system dependencies.

Additionally, there is a recognized gap in **data-driven operational insights** for vendors and organizations. Traditional coupon systems offer no analytical capabilities, and generic payment platforms do not provide vending-specific metrics such as peak beverage demand times, coupon redemption frequency, or user engagement patterns. TeaTech lays the groundwork for future analytical modules by storing structured transaction data, enabling meaningful insights through vendor dashboards or administrative tools in subsequent versions.

TeaTech's contributions can therefore be summarized as providing a **targeted, practical, and context-specific solution** that aligns digital transformation principles with the cultural and operational realities of hyper-local vending. By combining transparency, user validation, bilingual accessibility, and minimal adoption barriers with a lightweight technological footprint, TeaTech addresses the precise gaps left unfilled by existing systems. This makes it a significant and necessary innovation within the micro-vending domain.

CHAPTER 3: SYSTEM ANALYSIS AND REQUIREMENTS

3.1. Introduction

System analysis is a foundational phase in the software development lifecycle, serving as the bridge between problem identification and solution design. It involves examining the operational environment, understanding user needs, defining system objectives, and detailing the functional and non-functional requirements that guide implementation. For TeaTech, system analysis plays a particularly crucial role because the project aims to digitize an existing manual workflow without disrupting established financial and operational practices. This requires a thorough understanding of the interactions between employees, vendors, and existing coupon mechanisms to ensure that the digital system seamlessly integrates into real-world corporate vending environments.

The primary objective of TeaTech's system analysis is to translate the conceptual goals—such as transparency, error reduction, and enhanced user control—into structured requirements that can be implemented reliably. Since TeaTech is designed to operate in fast-paced corporate settings with high transaction density, the analysis phase must consider not only functional interactions but also constraints related to performance, accessibility, usability, and vendor adoption. These contextual factors directly inform the system's architecture, ensuring that the solution remains efficient, scalable, and user-friendly even under peak operational loads.

Another key aspect of the analysis involves identifying various user roles and mapping their respective workflows. TeaTech primarily supports two stakeholders: **employees**, who redeem digital coupons, and **vendors**, who initiate deduction requests. Each role interacts with the system differently and therefore requires dedicated interfaces, permission levels, and validation mechanisms. The analysis also considers potential future stakeholders—such as corporate administrators—whose needs may inform system expansion. Understanding these user dynamics helps define clear boundaries for what the system must accomplish in its initial implementation and where flexibility should be maintained for future enhancements.

TeaTech's analysis further incorporates real-world observations collected during the internship-based pilot testing phase. These observations reveal patterns of user behavior, common points of friction, and practical constraints that influence system design. For example, vendors require extremely rapid interactions to maintain service flow, while employees prioritize transaction visibility and confirmation control. Incorporating such insights ensures that system requirements are grounded in actual user needs rather than theoretical assumptions. This human-centered approach strengthens the solution's practical relevance and increases its likelihood of successful adoption.

Additionally, the analysis phase identifies the environmental conditions under which TeaTech must operate. Corporate beverage counters may experience variable network conditions, crowded spaces, and

high-frequency peak-hour usage. These factors shape requirements related to responsiveness, error handling, and interface simplicity. The system must support real-time updates, minimal latency, and intuitive workflows to function effectively in such dynamic settings. Any complexity beyond what users can manage during quick interactions would hinder adoption and reduce system effectiveness.

The analysis also delineates system boundaries by defining what TeaTech does **not** aim to address. Consistent with the project's scope, the analysis makes clear that TeaTech does not manage digital payments, vendor revenue reconciliation, or coupon issuance financial tracking. By documenting these exclusions explicitly, the system analysis ensures clarity of purpose and prevents scope creep during development. This disciplined approach helps maintain TeaTech as a lightweight and focused tool that solves a specific operational problem without introducing unnecessary complexities.

Finally, system analysis sets the stage for formalizing requirements in the subsequent sections. By integrating contextual understanding, user needs, environmental constraints, and project objectives, this chapter establishes the foundation for defining detailed functional and non-functional requirements. These specifications guide the design and implementation phases that follow, ensuring that every component of TeaTech aligns with its overarching mission of enhancing transparency, minimizing errors, and modernizing coupon redemption workflows within corporate settings.

3.2. Functional Requirements (FRs)

3.2.1. FR1: User Authentication

User authentication is a foundational functional requirement for TeaTech, ensuring that only authorized individuals gain access to the system's features and sensitive data. Since the platform manages digital coupon balances, transaction histories, vendor interactions, and bilingual interface settings, the integrity and confidentiality of user accounts are paramount. Authentication is not merely a preliminary step; it is a security-critical process that establishes trusted identities before enabling any operation within the system. In a corporate vending environment where both employees and vendors interact with the platform, a reliable authentication mechanism creates structured access control and prevents unauthorized use of the system.

The authentication requirement mandates that each employee and vendor must log into the system using unique credentials. These credentials typically include a registered email address or phone number along with a secure password. Incorporating a unique identifier ensures that every transaction—whether a coupon deduction, confirmation, or history query—is tied to a verifiable entity. This reduces ambiguity and strengthens accountability within the TeaTech workflow. By leveraging Firestore's integrated authentication services or equivalent secure identity frameworks, the system maintains strong safeguards for credential management and user verification.

Authentication also supports **role differentiation**, a necessary component for maintaining workflow integrity. Employees and vendors have distinct responsibilities within the system: employees approve or decline deduction requests, while vendors initiate them. Authentication ensures that a user logging in as

an employee cannot access vendor privileges, and vice versa. This separation of roles helps preserve operational boundaries and prevents misuse—intentional or accidental—of system features. The authentication layer thus forms the basis for implementing higher-level authorization rules.

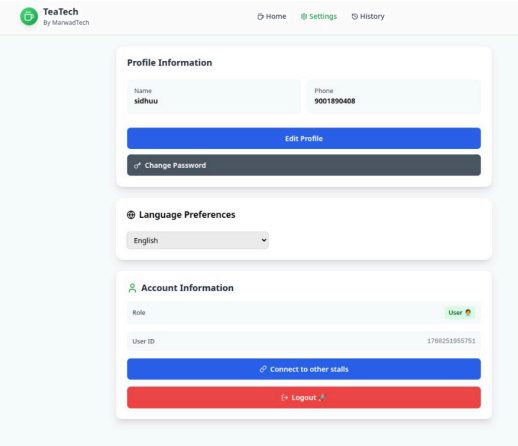
Another important element of the authentication requirement is the system's ability to manage **persistent sessions** securely. Users should remain logged in across multiple visits unless manually signed out, but session data must be protected against tampering or hijacking. This is particularly relevant in corporate settings where employees may use shared devices or vendors may operate from publicly accessible points. By implementing secure session tokens, periodic refresh mechanisms, and automatic expiration for inactive sessions, TeaTech reduces the risk of unauthorized access due to unattended devices.

To support usability, the authentication system also includes **mechanisms for account recovery**, such as password reset requests or email verification. In environments with diverse user demographics, these features reduce friction and ensure that users can regain access without vendor intervention or administrative support. Verification steps, such as confirming email ownership or validating an OTP, add a security layer against fraudulent recovery attempts. These capabilities contribute to a more resilient and user-friendly authentication ecosystem.

Furthermore, authentication plays a critical role in maintaining the **integrity of transaction logs**. Since each transaction is tied to an authenticated user ID, the system can generate accurate, tamper-resistant audit trails. This strengthens TeaTech's ability to resolve disputes, analyze usage patterns, and ensure transparency. Without authentication, logs would lose their reliability and traceability, undermining the platform's core objective of enhancing trust between users and vendors.

During pilot testing, authentication emerged as a natural fit for real-world workflows. Employees were comfortable signing in using their existing credentials, and vendors appreciated the system's simplicity in distinguishing roles. Authentication also streamlined subsequent system interactions by ensuring that each transaction carried a clear digital signature of its origin. These real-world insights affirm that strong yet user-friendly authentication is essential for TeaTech's operational success.

In summary, user authentication is a critical functional requirement that provides identity verification, role differentiation, session security, and reliable transaction traceability. It forms the backbone of TeaTech's digital workflow, ensuring that all interactions within the system are authorized, accountable, and securely managed. By implementing robust authentication mechanisms, TeaTech protects user data while enabling a seamless and trustworthy digital vending experience.



3.2.2. FR2: Real-time Balance Display

The real-time balance display is a core functional requirement of TeaTech, ensuring that users have immediate and accurate visibility into their remaining digital coupon credits at all times. Since coupon redemption is a continuous, high-frequency activity in corporate vending environments, users must be able to monitor their balance without delay or ambiguity. This requirement directly addresses the long-standing issues associated with paper coupon systems, where users often rely on memory, manual counting, or incomplete recollection to track coupon usage. By providing an instant and reliable balance display, TeaTech enhances user confidence, reduces disputes, and strengthens the transparency of the vending workflow.

The real-time balance feature is powered by Firebase Firestore's live synchronization capabilities, which update the user interface automatically whenever a balance change occurs. This means that the user does not need to refresh the page or perform any manual action to view the updated balance. The system responds to backend state changes instantly, reflecting deductions immediately after the user confirms a vendor's request. This smooth and dynamic interaction is essential for maintaining user trust, especially in peak-hour scenarios when rapid transactions occur back-to-back.

A key aspect of this requirement is **data consistency**. In multi-user environments where numerous employees may interact with multiple vendors simultaneously, it is critical that the displayed balance always reflects the actual stored value in the database. Firestore's atomic transaction model ensures that concurrent deduction attempts do not result in inconsistent or partial updates. For example, if two vendors mistakenly attempt to initiate deductions for the same user at the same time, the system maintains a single source of truth and prevents race conditions that could otherwise lead to incorrect balances. This level of consistency is essential for operational reliability.

The real-time balance display also contributes to **user empowerment**, allowing employees to make informed decisions about their coupon usage. When users can instantly verify their balance after each transaction, they feel more in control of the redemption process. This reduces anxiety around unexpected deductions and eliminates the guesswork that was prevalent in traditional coupon systems. Furthermore, because TeaTech logs every transaction, the real-time balance is complemented by a detailed transaction history, enabling users to cross-check previous deductions whenever necessary.

From a usability perspective, the balance display must be **clear, prominent, and accessible**. The system interface presents the balance at the top of the user dashboard or home screen, formatted in a simple and readable manner. This ensures that users can check their balance at a glance without navigating through complex menus. During pilot testing, users expressed appreciation for this immediacy, noting that it simplified interactions with vendors and removed the need to mentally track remaining coupons throughout the day.

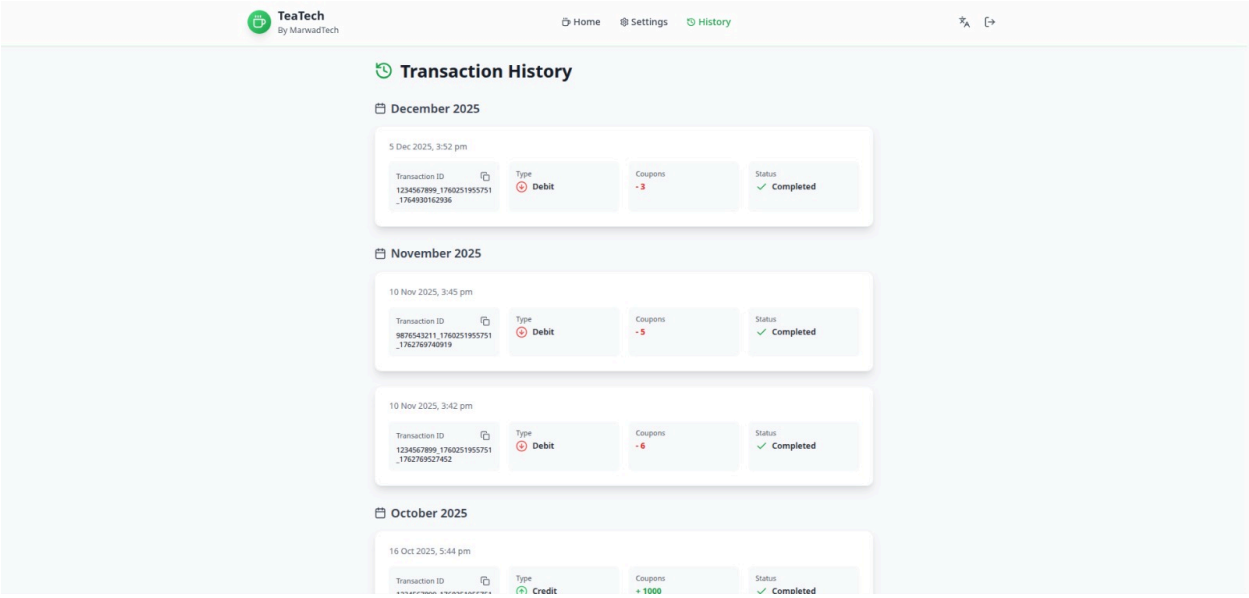
The real-time display requirement also plays a significant role in **supporting the two-step confirmation workflow**. When the vendor initiates a deduction request, the user sees not only the pending request but

also the anticipated balance after approval. This predictive visibility allows the user to validate the accuracy of the request before confirming it. If the request appears incorrect, the user can decline it, preventing errors before they affect the balance. This dual-layer verification is a substantial improvement over paper systems, where such real-time comparisons are impossible.

Additionally, the real-time balance feature assists vendors in their operations. Although the primary display is on the employee side, vendors can indirectly trust the correctness of user balances because deductions occur only after digital confirmation and database synchronization. This removes any reliance on employee claims about remaining coupons and eliminates the mismatches that commonly occur in manual workflows. Vendors can serve customers more efficiently knowing that the system enforces accuracy automatically.

From a technical standpoint, the requirement ensures that the balance display remains functional under varying network conditions. Although live updates depend on internet connectivity, Firestore's offline persistence enables the last-known balance to remain visible even when the connection drops temporarily. Once connectivity is restored, the system seamlessly synchronizes updates. This hybrid behavior ensures robustness in environments where network strength fluctuates, such as semi-open beverage counters.

In summary, the real-time balance display is a vital functional requirement that enhances transparency, minimizes disputes, and supports seamless interaction between users and vendors. By offering instant updates, ensuring data consistency, integrating with confirmation workflows, and providing intuitive UI visibility, TeaTech significantly improves the reliability and user experience of corporate vending systems. This feature stands as one of the most important differentiators between TeaTech and traditional paper-based coupon workflows.



3.2.3. FR3: Vendor Linking

Vendor linking is a fundamental functional requirement in TeaTech, ensuring that employees can interact only with authorized vendors within the corporate environment. This mechanism establishes a structured relationship between users and vendors, enabling the system to determine which vendor is permitted to initiate coupon deduction requests for a given employee. By formalizing this linkage, TeaTech prevents unauthorized vendors from accessing or manipulating employee balances, thereby reinforcing the system's security, integrity, and operational clarity.

At its core, vendor linking serves two critical purposes. First, it defines the **scope of interaction** between employees and vendors. In many corporate settings, multiple beverage counters or tea stalls may operate simultaneously, each with its own vendor. Without structured linking, the system would have no way to distinguish which vendor a user is associated with or prevent unauthorized deduction attempts. By establishing explicit links, TeaTech ensures that request initiation is restricted to designated vendors, maintaining proper boundaries within the workflow.

Second, vendor linking supports **operational scalability**. As organizations grow, new vendors may join or existing vendors may operate across multiple locations. TeaTech's linking mechanism allows administrators—or the system during onboarding—to assign vendors to users dynamically based on workplace layout, shift timings, or organizational structure. This flexibility supports deployments in small offices, large IT parks, and multi-vendor corporate campuses without requiring major system reconfiguration.

The linking process typically involves assigning each user a reference identifier associated with a specific vendor. This mapping is stored in the database and used by the system to validate deduction requests. When a vendor initiates a transaction, the system cross-checks whether the vendor ID matches the user's assigned vendor. Only if the match is valid does the system allow the request to proceed to the user for confirmation. This prevents users from receiving deduction requests from vendors they do not frequent or from vendors attempting to deduct mistakenly or maliciously.

Vendor linking also enhances **transaction traceability**, as each interaction is tagged with a vendor identifier. This allows employees to view their history with complete contextual clarity—knowing which vendor initiated each deduction and when. In case of disputes, these logs become invaluable for reconciling differences or identifying workflow anomalies. Vendors benefit as well, as they can access filtered histories showing only the transactions they initiated, enabling accurate end-of-day or end-of-month reconciliation.

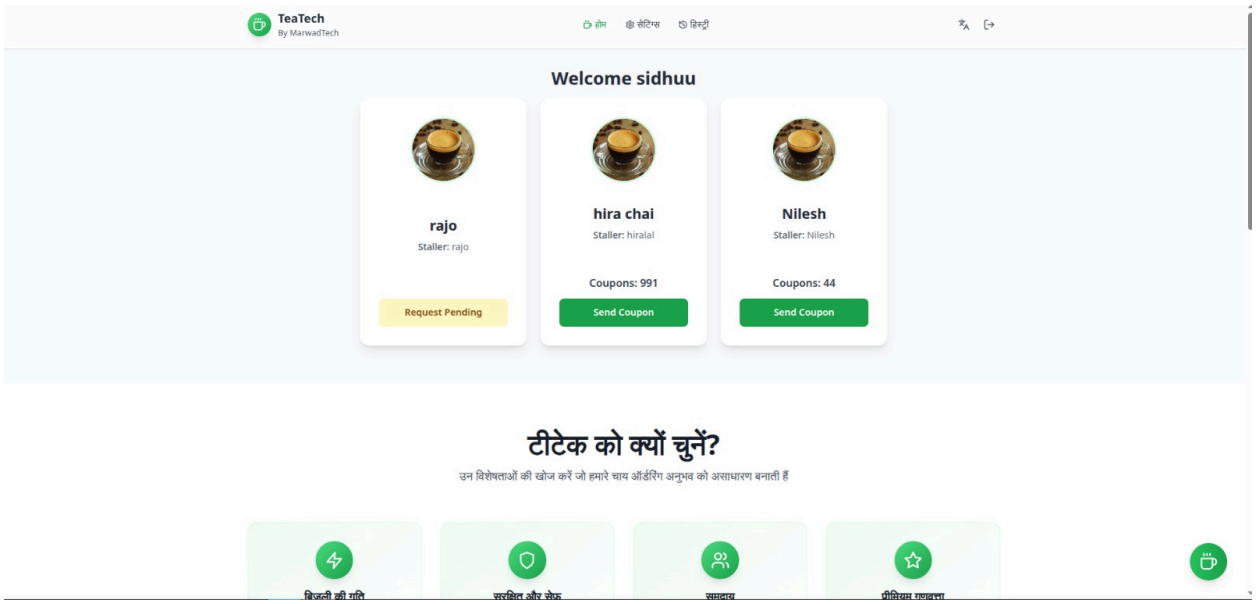
Additionally, the vendor linking framework supports **multi-vendor environments**, which are common in large corporate campuses. An employee may regularly visit a specific vendor due to convenience or preference, and the system facilitates this relationship seamlessly. At the same time, if an employee needs to interact with a different vendor temporarily—such as during maintenance, shift changes, or vendor unavailability—the system can support dynamic relinking or multi-vendor permissions with minimal configuration effort. These capabilities ensure that TeaTech remains flexible without compromising security.

Vendor linking plays a crucial role in **preventing unauthorized activity**. Without linking, any vendor could theoretically attempt to initiate deductions for any user. Even though TeaTech's two-step confirmation mechanism would still require user approval, the presence of unsolicited requests could create confusion, mistrust, or operational disruption. By filtering requests at the system level before they reach the user, vendor linking eliminates unnecessary noise and maintains a smooth user experience.

During pilot testing, vendor linking significantly improved the predictability and efficiency of interactions. Employees appreciated that they received deduction requests only from their designated vendor, while vendors valued the clarity and control provided by the system. This structure allowed vendors to process customers more efficiently without sifting through irrelevant or misdirected requests. It also helped eliminate accidental interactions that could occur in high-traffic corporate settings.

From a technical perspective, vendor linking requires careful data modeling to ensure scalability and performance. The system must maintain accurate mappings between users and vendors while ensuring quick lookup times during request initiation. Firestore's fast document queries and indexing capabilities support this requirement, enabling the system to perform validation checks without introducing noticeable latency. This ensures that linking contributes positively to the overall user experience and does not slow down transaction workflows.

In summary, vendor linking is an essential functional requirement that establishes authorized interaction channels, enhances security, improves auditability, and supports scalability within TeaTech. By defining clear user–vendor relationships, the system prevents unauthorized deductions, maintains operational order, and ensures that all interactions occur within a controlled and trusted framework. This requirement plays a foundational role in ensuring that TeaTech remains both user-friendly and secure in real-world corporate deployments.



3.2.4. FR4: Two-Step Request Interface

The two-step request interface is a central functional requirement of TeaTech, serving as the structural foundation for the system's secure and transparent coupon deduction workflow. This interface governs the interaction between vendors and employees by ensuring that a deduction request must pass through two clearly defined stages: **initiation by the vendor** and **explicit approval by the user**. The design of this interface directly addresses critical challenges identified during problem analysis, including miscommunication, accidental deductions, and disputes arising from ambiguous or unverifiable paper coupon exchanges.

The first stage of the interface begins with the **vendor-initiated request**. When a user arrives at the beverage counter and indicates the beverage they wish to purchase, the vendor inputs the number of coupons to be deducted into the system. This initiation sends a structured digital request to the employee's interface, containing essential metadata such as vendor identity, requested quantity, and timestamp. Instead of immediately deducting the coupon balance, the system marks the request as "pending" and awaits the user's response. This ensures that no irreversible action is taken without user participation.

The second stage involves the **user-facing confirmation screen**, where the employee reviews the pending request. The interface presents the request in a clean and readable format, allowing the user to verify the correctness of the vendor's input. The user then chooses to either **approve** or **decline** the request. If approved, the system processes the deduction and updates both the vendor's and the user's interfaces in real time. If declined, the request is canceled, and the vendor receives feedback indicating that the user did not authorize the transaction. This dual-validation mechanism eliminates ambiguity and ensures that both parties are consciously involved in the transaction flow.

A core strength of the two-step request interface lies in its ability to **prevent accidental or unauthorized deductions**. In traditional systems, vendors may unintentionally mark or collect coupons incorrectly due to distractions, high footfall, or rushed interactions. The two-step interface removes the risk of unilateral action by requiring user approval. Even if a vendor mistakenly initiates an incorrect quantity, the user can catch and decline the request before any balance change occurs. This creates a built-in error prevention mechanism that aligns with real-world vending dynamics.

Another benefit of this interface is its role in reducing **verbal dependency**. In manual workflows, deductions often rely on spoken confirmation ("One tea? One coupon?") which can be easily misunderstood in crowded or noisy environments. TeaTech replaces these informal cues with an unambiguous digital prompt that removes subjective interpretation. This ensures that service remains efficient even during peak hours when clarity is essential but difficult to maintain through verbal exchanges alone.

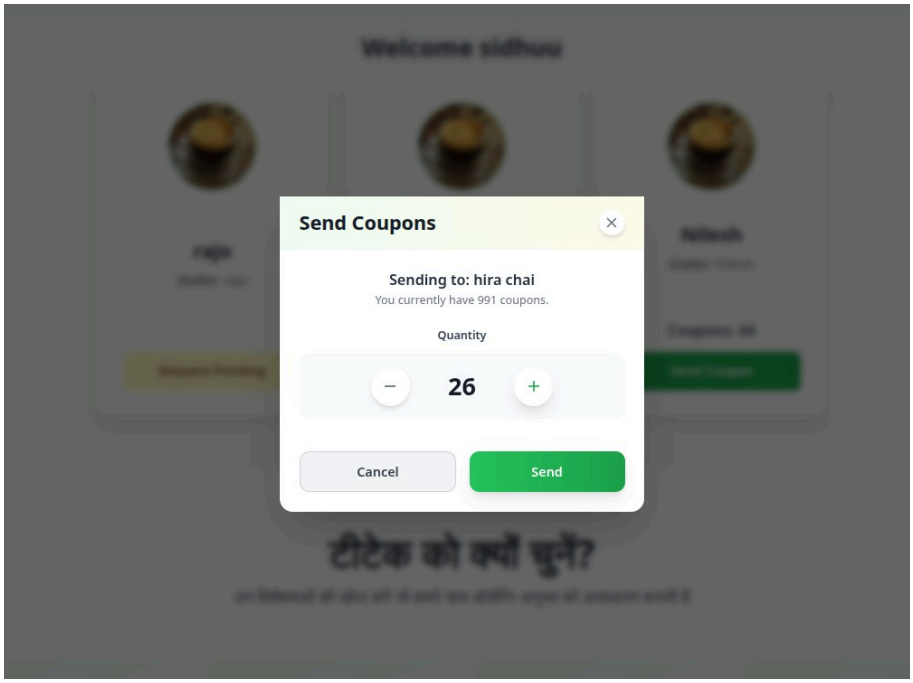
From a user experience perspective, the two-step interface is designed to be **minimalistic, intuitive, and fast**. The confirmation prompt appears prominently on the user's screen and can be approved with a single tap, ensuring that the workflow does not slow down the vending process. Pilot testing showed that employees appreciated this interface because it offered control without introducing complexity. Vendors similarly found that the interface reduced the likelihood of disputes, as users were directly verifying each deduction.

The two-step request interface also reinforces **auditability**, as each step of the interaction is recorded in the database. The initiation event, the user's decision, the timestamp, and the final deduction outcome are all stored as part of a structured transaction log. This level of detail provides comprehensive traceability, enabling both employees and vendors to revisit past interactions whenever reconciliation or clarification is required. The system thus transforms an informal exchange into a fully documented digital process.

Technically, implementing the two-step interface requires efficient real-time communication between user and vendor devices. Firebase Firestore's real-time listeners allow the system to instantly detect when a vendor sends a request and push the confirmation prompt to the user's screen without delay. Similarly, the vendor's interface updates automatically when the user approves or declines the request. Maintaining this seamless back-and-forth communication ensures that the two-step process does not hinder service flow.

Additionally, the interface accommodates scenarios where multiple pending requests may exist, such as when a user interacts with a vendor who mistakenly sends duplicate requests. The system displays such occurrences in a clear queue-like structure, allowing the user to respond selectively or ignore irrelevant requests. This prevents interface clutter and ensures operational reliability even in edge cases.

In summary, the two-step request interface is an essential functional requirement that ensures security, reduces errors, enhances transparency, and strengthens user trust. By transforming a traditionally informal and error-prone interaction into a structured digital workflow, TeaTech significantly improves the accuracy and efficiency of corporate vending systems. This interface embodies the system's user-centered design philosophy and plays a pivotal role in differentiating TeaTech from both traditional coupon methods and generic digital platforms.



3.2.5. FR5: Transaction Confirmation

Transaction confirmation is a critical functional requirement that finalizes the coupon redemption workflow within TeaTech. While the two-step request interface governs the interaction leading up to a deduction, transaction confirmation formalizes the moment at which the system validates the user's decision and updates the digital state. This requirement ensures that every deduction is authorized, securely recorded, and reflected in real time across all relevant interfaces. By making the confirmation step explicit and mandatory, TeaTech closes the loop between vendor initiation and system-level enforcement, guaranteeing transactional integrity.

At the core of this requirement is the **user's explicit approval action**, which serves as the authoritative trigger for balance deduction. The system waits for the employee to press "Approve" before performing any database updates. This action reflects the user's acknowledgement that the vendor's request is correct and intended. If the user selects "Decline," the system immediately cancels the request without modifying the coupon balance. This binary design removes ambiguity and ensures that the system never assumes user consent without receiving it clearly.

Once a transaction is approved, the system performs an **atomic deduction operation** on the user's digital coupon balance. Atomicity, ensured through Firestore's transactional writes, guarantees that the deduction either completes fully or does not occur at all. This prevents errors that could arise from concurrent requests, network delays, or device interruptions. For example, if two deduction requests are mistakenly or simultaneously sent for the same user, the system processes them one at a time, maintaining the integrity of the stored balance. This protects users from double deductions and vendors from unintentional undercounting.

After a successful deduction, TeaTech triggers **real-time interface updates**. The user's balance display immediately reflects the new value, and the vendor's interface shows that the request was successfully completed. This instant synchronization is essential for maintaining a seamless vending experience, especially in environments where rapid customer turnover is common. Without real-time updates, users might feel uncertain whether a deduction occurred, and vendors might unintentionally initiate redundant requests. The confirmation step thus ensures clarity for both participants and reduces operational delays.

Another important component of transaction confirmation is the creation of a **permanent audit log entry**. Each confirmed transaction is recorded with key metadata such as timestamp, user identifier, vendor identifier, coupon quantity, and transaction status. This log is central to TeaTech's goal of eliminating disputes. In case of disagreement, both employee and vendor can review the exact time and details of the transaction. This level of transparency is impossible in traditional paper coupon systems and represents one of TeaTech's most meaningful improvements.

Transaction confirmation also supports **error prevention** by enabling users to verify the vendor's request before the final deduction. If a vendor inputs the wrong number of coupons or selects the wrong user, the confirmation prompt provides an opportunity to catch the error before it affects the coupon balance. This prevents accidental deductions—an issue frequently observed in manual workflows—and reduces the likelihood of disputes. In pilot testing, users consistently reported that this ability to review and confirm deductions increased their trust in the system.

From a user experience perspective, transaction confirmation is designed to be **fast, intuitive, and unobtrusive**. The confirmation prompt appears clearly on the user's screen, requiring only a single tap to approve or decline. This ensures that the redemption process remains smooth and does not slow down service. The clarity of this step also minimizes cognitive load on the user, allowing them to interact confidently even in rushed or noisy environments.

For vendors, transaction confirmation provides operational reassurance. Because the system notifies vendors immediately when a transaction is confirmed or declined, they no longer rely on verbal feedback or assumptions. This minimizes miscommunication and ensures that beverage service continues smoothly. Vendors can quickly identify whether they should proceed with serving a beverage, wait for user confirmation, or cancel and retry a request. This degree of coordination significantly reduces friction at vending counters.

Finally, transaction confirmation ensures **system reliability** across variable network conditions. Even if connectivity fluctuates during the workflow, the system processes confirmations as soon as a stable connection is restored, maintaining consistent behavior without duplicating or losing requests. Firestore's synchronization model helps preserve data integrity in such edge cases.

In summary, transaction confirmation is a crucial functional requirement that finalizes and safeguards every coupon deduction. By enforcing explicit user approval, ensuring atomic balance updates, generating real-time feedback, and maintaining comprehensive audit logs, this requirement strengthens operational accuracy and trust. It transforms a potentially ambiguous interaction into a transparent, secure, and verifiable digital transaction—aligning perfectly with TeaTech's mission.

3.2.6. FR6: Transaction History

Transaction history is a core functional requirement of TeaTech, providing users and vendors with a comprehensive, tamper-proof record of all coupon-related activities. This feature directly supports TeaTech's central objective of enhancing transparency and eliminating disputes that arise from ambiguous or unverifiable transactions. By maintaining a detailed log of every confirmed deduction, the system creates a persistent and trustworthy data trail that neither traditional paper coupons nor informal verbal confirmations can offer. Transaction history serves as both a user empowerment tool and an operational safeguard, ensuring accountability across the entire vending workflow.

The transaction history module records key metadata for every deduction event, including the **timestamp**, **vendor identity**, **user identity**, **quantity of coupons deducted**, and **transaction status**. This structured collection of information enables users to trace their coupon usage with precision. When employees want to verify how many coupons they have consumed over a given period—or identify when a specific deduction took place—they can simply refer to their personal transaction log. This functionality replaces the unreliable process of manually remembering or estimating coupon usage, which often leads to misunderstandings in paper-based systems.

For vendors, transaction history provides an equally valuable resource. Vendors can review all deductions they have initiated, allowing them to reconcile daily or monthly coupon collections with ease. Since the logs are automatically generated and stored by the system, vendors are relieved of the burden of

maintaining handwritten records or manually counting stacks of paper coupons. This improves operational accuracy and reduces the administrative effort required to manage high volumes of transactions. In multi-vendor environments, each vendor's history remains isolated to their own operations, ensuring clarity and preventing confusion.

A critical advantage of TeaTech's transaction history is its role in **dispute resolution**. In traditional coupon workflows, disagreements frequently occur because neither party possesses reliable evidence of past interactions. TeaTech addresses this challenge directly by offering indisputable, system-generated logs. If an employee believes a deduction was inaccurate, or if a vendor suspects undercounting, both can refer to the transaction history to identify the exact event in question. This eliminates subjective debates and fosters an environment of trust and accountability.

The transaction history module also enhances **user awareness and control**. Users can review their consumption patterns, track how quickly their coupon balance is depleting, and plan future purchases accordingly. This visibility is especially beneficial in settings where employees buy coupons in bulk and use them gradually over several days or weeks. The history acts as a transparent ledger that supports informed decision-making and reduces anxiety about unexpected balance fluctuations.

From a design perspective, the transaction history interface must be **intuitive, readable, and accessible**. Each entry should be clearly formatted, with key data points highlighted to ensure quick interpretation. Filters or sorting options may be included to help users navigate large histories efficiently, particularly in long-term usage scenarios. Although the initial implementation of TeaTech focuses on basic listing functionality, the system architecture allows for future enhancements such as date filtering, monthly summaries, or export capabilities.

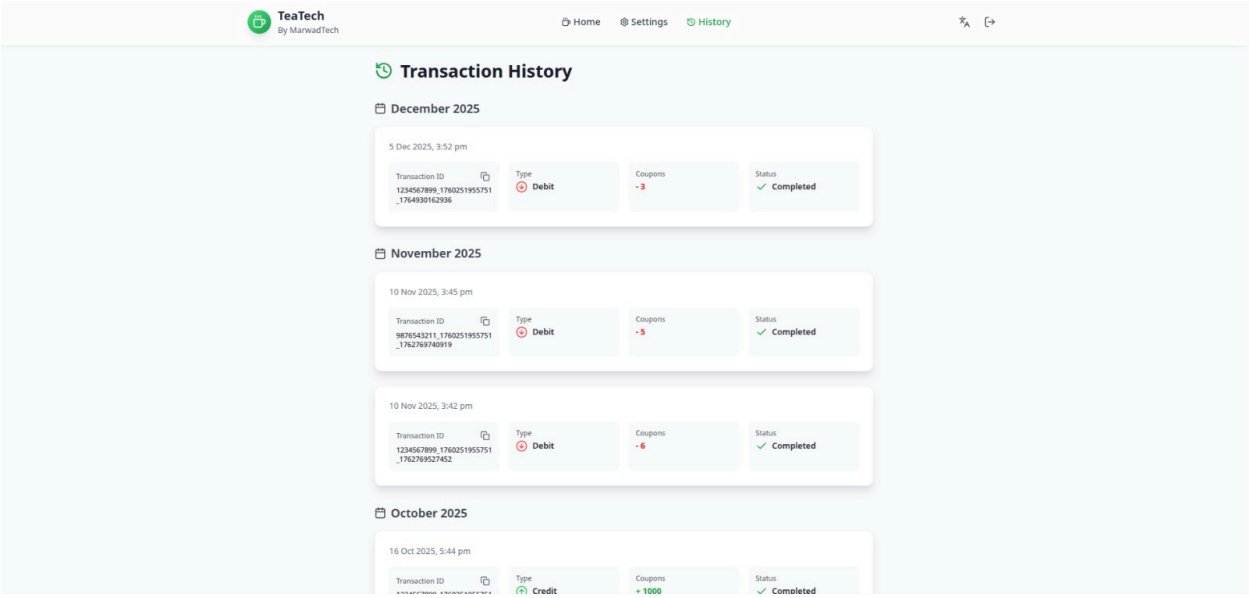
Another important aspect of the requirement is the **immutability** of transaction logs. Once a transaction is confirmed and recorded, it cannot be edited or deleted by users or vendors. This tamper-proof design preserves the integrity of the audit trail. Any attempt to manipulate records would undermine TeaTech's commitment to transparency and could reintroduce mistrust. Firestore's structured document storage and security rules help enforce this immutability, ensuring that logs remain accurate and unaltered.

Transaction history also contributes to **analytics potential**. Although advanced analytics are not part of the initial scope, the availability of historical data enables future development of vendor performance dashboards, consumption trend analysis, peak demand forecasting, and organizational insights. Storing granular transaction data from the outset ensures that TeaTech is prepared for such expansions, aligning with its long-term scalability goals.

During pilot testing, transaction history emerged as one of the most appreciated features among employees. Users expressed that having a digital record provided reassurance, especially in situations where their remaining balance differed from what they expected. Vendors similarly valued the clarity it provided, reducing reliance on memory or rough estimates. This real-world validation underscores the importance of including a robust history module as a core requirement.

In summary, transaction history is an essential functional requirement that enhances transparency, strengthens accountability, and supports operational reliability within TeaTech. By offering a complete

and immutable record of every coupon deduction, the system eliminates ambiguity, improves trust between users and vendors, and lays the groundwork for future analytical capabilities. This feature significantly elevates TeaTech beyond traditional coupon workflows, transforming informal micro-transactions into structured, auditable digital events.



3.2.7. FR7: Bi-lingual Support

Bi-lingual support is an important functional requirement of TeaTech, enabling the platform to be accessible and user-friendly for a diverse range of stakeholders within corporate environments. Corporate beverage counters often serve employees from various linguistic backgrounds and are commonly operated by local vendors who may not be comfortable navigating English-only interfaces. By offering TeaTech in both English and Hindi, the system ensures equitable usability, promotes inclusivity, and removes language-related barriers that could hinder adoption. This requirement reflects a deep understanding of the socio-cultural context in which TeaTech operates and underscores the system's commitment to practical usability.

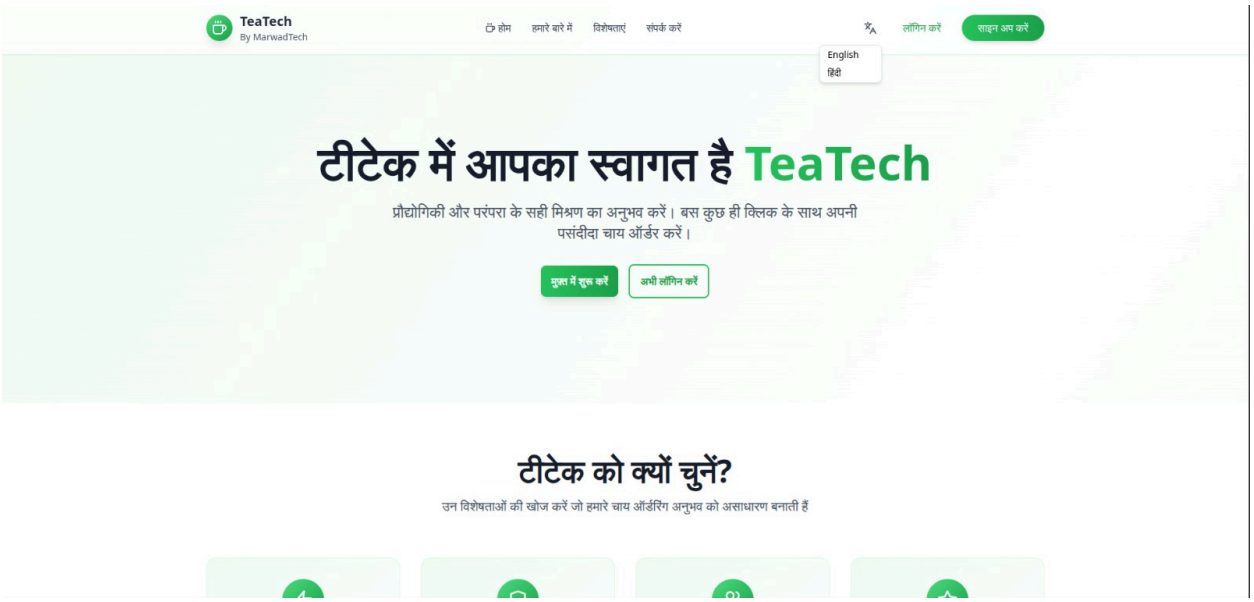
The need for bilingual support emerges from real-world observations during the project's pilot stage. While employees in corporate offices generally possess high levels of digital literacy, many beverage vendors rely on simple, intuitive interfaces and may prefer operating in their native language. An English-only system could introduce confusion, increase cognitive load, or lead to mistakes in initiating or interpreting deduction requests. By incorporating Hindi—a widely spoken language across India—TeaTech accommodates these users directly, ensuring that vendors can interact confidently with the system without requiring external assistance or translation.

From a user interface perspective, bilingual support requires the system to dynamically adjust all text elements based on the selected language. This includes buttons, labels, prompts, confirmation dialogs, transaction logs, error messages, and navigation menus. The system must ensure that both language versions offer a consistent experience, maintaining clarity and visual balance across the interface. This

consistency is crucial because inconsistencies can cause misunderstandings or misinterpretations during transactions, especially when users switch between languages.

The implementation of bilingual support also plays a vital role in **reducing operational errors**. Consider a scenario in which a vendor must read a confirmation status or interpret a system instruction quickly during peak service hours. If the message appears in an unfamiliar language, the vendor may misinterpret the request, leading to incorrect deductions or user confusion. By allowing vendors to choose the language they are most comfortable with, TeaTech minimizes the risk of such errors and enhances workflow stability. This feature becomes especially important in environments where service must be fast, accurate, and uninterrupted.

TeaTech's bilingual requirement extends beyond interface text to include **system-generated notifications and prompts**, such as pending request alerts or approval messages. Both employees and vendors must clearly understand the meaning of each notification to avoid miscommunication. Ensuring high-quality translations that reflect everyday language rather than overly technical terminology helps the system remain intuitive and relatable for users. Additionally, consistent terminology across the interface reinforces user familiarity and reduces learning curves.



During pilot deployment, bilingual support proved instrumental in achieving smooth onboarding for vendors. Vendors expressed that using the system in Hindi allowed them to operate independently without relying on supervisors or employees for guidance. This autonomy improved their confidence and reduced friction at the beverage counter. Employees also appreciated that the system accommodated vendor preferences without affecting their own experience, since they could choose their preferred language independently.

In summary, the bilingual support requirement strengthens TeaTech's usability, inclusivity, and operational effectiveness. By offering the platform in both English and Hindi, TeaTech reduces linguistic barriers, minimizes errors, and ensures that all users—regardless of language preference—can engage

with the system confidently. This feature exemplifies TeaTech's commitment to contextual relevance and user-centered design, making it a practical and widely adoptable solution for corporate vending environments.

3.3. Non-Functional Requirements (NFRs)

3.3.1. Performance (Latency, Scalability)

Performance is a critical non-functional requirement for TeaTech, ensuring that the system remains responsive, efficient, and reliable under varying load conditions. Given that the platform is deployed in fast-paced corporate vending environments with high-frequency, low-latency interactions, it must deliver rapid feedback and maintain consistent behavior regardless of the number of simultaneous users. Performance requirements encompass two primary dimensions: **latency**, which concerns the speed of system operations, and **scalability**, which pertains to the system's ability to handle growing workloads without degradation.

Latency Requirements

In corporate beverage counters, transactions occur in rapid succession, often during peak hours when employees gather for breaks. Under such circumstances, any delay in processing vendor requests, displaying confirmation prompts, or updating user balances can disrupt the flow of service. To maintain operational efficiency, TeaTech must ensure **sub-second response times** for core interactions such as:

- Receiving and displaying vendor deduction requests
- Updating user balances after confirmation
- Reflecting real-time transaction statuses
- Synchronizing data across devices

The system relies on Firebase Firestore's real-time listeners to achieve low-latency performance. Firestore pushes updates instantly to connected clients, bypassing the need for user-initiated refreshes. This real-time synchronization ensures that deduction requests appear immediately on employee devices and that vendors receive instant feedback after approval. Such responsiveness is essential for maintaining user trust and preventing congestion at vending counters.

Latency performance also includes minimizing delays caused by network instability. While TeaTech requires an internet connection to process confirmations, Firestore's offline persistence helps mitigate disruptions by caching data locally when connectivity drops. Once the connection is restored, updates are synchronized seamlessly. This hybrid behavior allows the system to remain functional even in environments with fluctuating network strength.

Scalability Requirements

Scalability ensures that TeaTech remains robust as the system grows—whether due to increasing users, vendor expansion, or deployment across multiple corporate sites. A scalable architecture is crucial because the system must accommodate:

- High transaction volumes during peak breaks
- Multiple vendors operating concurrently
- Hundreds or thousands of employees interacting daily
- Real-time updates exchanged across potentially large device networks

Firestore's NoSQL database model supports horizontal scaling, allowing the system to handle large datasets and concurrent read-write operations without performance degradation. Its document-based structure enables efficient indexing and querying, preventing bottlenecks even when transaction logs grow over time. The stateless nature of the frontend further improves scalability by keeping the client interface lightweight and delegating heavy lifting to the database backend.

Additionally, TeaTech's architectural separation of roles—employee, vendor, and potentially administrators in the future—ensures that scaling one dimension does not burden others. For example, increasing the number of vendors only requires adding new vendor profiles rather than restructuring the entire system. This modularity contributes to the system's long-term adaptability.

Performance requirements also include **handling concurrency**. In scenarios where multiple vendors attempt to interact with the same user or where users process multiple pending requests, Firestore's atomic transactions ensure that data remains consistent. Concurrent operations do not overwrite each other or create partial updates, preserving system integrity under load.

Operational Impact of Performance Requirements

A high-performing system improves user satisfaction, vendor efficiency, and overall workplace productivity. Low latency reduces queue buildup, ensuring employees spend minimal time at beverage counters. Scalable performance enables organizations to adopt TeaTech across multiple departments or campuses without requiring architectural redesigns. Moreover, predictable and stable performance supports the reliability expected in corporate environments where hundreds of micro-interactions occur daily.

During pilot testing, performance emerged as a key determinant of user acceptance. Employees responded positively to instant balance updates and rapid confirmation interactions. Vendors reported that the system did not slow down service, even during peak hours, and appreciated the absence of loading delays. These real-world observations validated the importance of achieving stringent performance standards in the system's non-functional design.

In summary, performance—through optimized latency and robust scalability—is essential to TeaTech's success. By utilizing Firestore's real-time capabilities, adopting a scalable data structure, and ensuring consistent low-latency interactions, TeaTech delivers a seamless, responsive, and efficient user experience that supports high-volume vending operations without compromise.

3.3.2. Security (Data Integrity, Authentication, Availability)

Security is a critical non-functional requirement for TeaTech, as the system manages sensitive transactional data, user identities, vendor interactions, and coupon balances that must remain accurate, confidential, and accessible only to authorized parties. Given that TeaTech operates in a corporate environment where trust and transparency are essential, the security framework must ensure that the system resists unauthorized access, prevents data corruption, and remains consistently available during peak operational hours. The three major pillars of TeaTech's security model—data integrity, authentication, and availability—collectively uphold the platform's reliability and protect its users from operational and financial discrepancies.

Data Integrity

Data integrity refers to the accuracy, consistency, and immutability of information maintained within the system. Since TeaTech tracks coupon balances and transaction histories, even a minor data inconsistency can lead to disputes or mistrust between employees and vendors. The system mitigates these risks through Firestore's **atomic transactions**, which guarantee that coupon deductions either complete fully or not at all. This prevents conditions such as partial updates, double deductions, or race conditions that can occur when multiple interactions target the same data simultaneously.

Additionally, the system enforces strict **validation rules** at the database level. Firestore security rules restrict write operations so that only authenticated users and authorized vendors can update specific documents. Unauthorized users cannot manipulate data, delete logs, or modify balances. Furthermore, all transaction logs are **append-only**, ensuring that no transaction can be modified or erased once recorded. This immutability is essential for maintaining an auditable system that can withstand scrutiny during dispute resolution or organizational audits.

Authentication

Authentication ensures that only legitimate users—employees and vendors—can access TeaTech's functionalities. By requiring secure login credentials and role-based identity verification, the system prevents unauthorized parties from accessing sensitive information or initiating fraudulent deductions. Authentication measures include:

- **Unique user identities** tied to email or phone numbers
- **Secure passwords** following recommended complexity standards
- **Session management** using secure tokens that expire automatically after inactivity
- **Protection against impersonation**, ensuring vendors cannot log in as users and vice versa

These mechanisms collectively ensure that every interaction in the system originates from a verified and trusted user. Authentication directly supports transactional security as well: every deduction is tagged with a unique user and vendor identity, making all actions traceable and verifiable.

Availability

Availability ensures that TeaTech remains operational and responsive even under varying network conditions, high user loads, or localized failures. Since beverage counters serve large volumes of employees during short intervals, system downtime can create bottlenecks and disrupt service. TeaTech leverages Firestore's highly distributed infrastructure to provide **high availability**, reducing the risk of outages that might affect transaction processing.

The system also incorporates **offline caching** on user devices. If temporary network disruptions occur, users can still view their most recent cached balance and transaction history. Once connectivity is restored, the system synchronizes changes seamlessly. This hybrid approach enhances reliability in semi-open cafeteria environments where network stability may fluctuate.

To further support availability, TeaTech's frontend is built as a lightweight web client with minimal dependencies. This reduces load times, minimizes failure points, and ensures the interface remains responsive even on low-powered vendor devices.

Security Impact on User Trust

The combination of data integrity, authentication, and availability directly influences user confidence. Employees trust that their balances are accurate, that no unauthorized deductions can occur, and that their usage history is preserved. Vendors rely on the system to maintain consistent counts, prevent tampering, and support operational continuity. This secures TeaTech's position as a reliable alternative to paper-based coupon workflows, addressing core problems of disputes, mistrust, and ambiguity.

3.3.3. Usability (UX, Localization)

Usability is a vital non-functional requirement that ensures TeaTech remains intuitive, accessible, and efficient for all types of users—including employees, vendors, and administrators in future iterations.

Since TeaTech is deployed in fast-paced corporate vending environments, the system must accommodate varying levels of digital familiarity, linguistic diversity, and high-frequency interactions. The usability requirement is anchored in two core principles: **user experience (UX)** and **localization**, both of which are essential for maximizing adoption and minimizing operational friction.

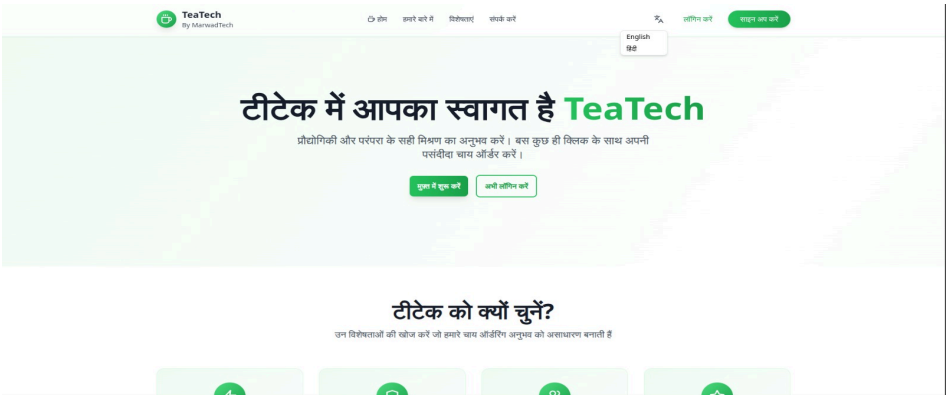
User Experience (UX)

The user experience component focuses on creating an interface that is simple, responsive, and easy to navigate. TeaTech must support users who interact with the system multiple times a day, often in hurried or crowded settings. As such, the design emphasizes clarity, speed, and minimalism. Key UX considerations include:

- **Clear balance visibility** on the main dashboard, ensuring users can instantly view their remaining coupons.
- **Prominent confirmation prompts** that display vendor requests with large, readable buttons for approving or declining.
- **Consistent iconography and color coding** to differentiate between pending, completed, and canceled transactions.
- **Low cognitive load**, ensuring no unnecessary steps or complex navigation is required to complete everyday actions.

UX design also accounts for environmental constraints. At beverage counters, users may be distracted by noise, standing position, or limited time. Interface elements are therefore optimized for rapid interpretation, reducing the chances of mistakes such as accidental approvals.

For vendors, the UX requirement is equally important. Vendor interfaces must remain extremely lightweight and responsive so that the process of generating deduction requests does not interrupt service flow. High accuracy in request initiation, clear notifications upon user responses, and minimal navigation complexity ensure a smooth vendor experience even during rush hours.



Localization

Localization extends TeaTech's usability by making the system linguistically accessible to its diverse user base. Corporate campuses in India commonly employ vendors who prefer regional languages, and employees may also span various linguistic backgrounds. To accommodate this, TeaTech implements bilingual support, enabling seamless switching between **English** and **Hindi**.

Localization involves more than just translating words. It requires adapting user interfaces to ensure that text lengths, phrasing, instructional elements, and cultural nuances are preserved across languages. This ensures uniform clarity regardless of linguistic choice. The localization strategy includes:

- **JSON-based translation files** that store key-value pairs for UI text.
- **Dynamic rendering**, allowing the interface to update instantly when users switch languages.
- **Consistent terminology** across all interface components, ensuring users are not confused by varying phrases or labels.

Pilot feedback confirmed the effectiveness of localization: vendors preferred interacting with the system in Hindi, while employees appreciated having the option to choose based on comfort. This dual-language flexibility significantly reduces onboarding friction and promotes widespread adoption.

Usability Impact on Adoption

Strong usability ensures that TeaTech integrates smoothly into existing workflows without requiring extensive training or technical support. A well-designed user experience encourages daily use, reduces error rates, and increases overall system satisfaction. Likewise, localization enhances inclusivity, ensuring that no user—regardless of language ability—feels excluded from the digital workflow.

By prioritizing UX and localization as non-functional requirements, TeaTech confidently addresses the human factors that influence the success of digital systems. These considerations ensure that the platform remains approachable, culturally compatible, and operationally efficient in real-world corporate vending environments.

CHAPTER 4: SYSTEM DESIGN

4.1. System Architecture

4.1.1. Presentation Tier (Frontend)

The Presentation Tier forms the user-facing layer of TeaTech, responsible for delivering a clear, intuitive, and responsive interface to employees and vendors. As the first point of interaction in the system architecture, the frontend plays a critical role in ensuring that the digital coupon workflow remains accessible, efficient, and easy to understand even during fast-paced vending operations. Built using HTML5, modern JavaScript (ES6+), and Tailwind CSS, the frontend is designed to balance visual simplicity with functional robustness, allowing users of varying technical proficiency to navigate the system with confidence.

The central objective of the Presentation Tier is to abstract technical complexity and present only the information and actions that users need in any given moment. For employees, this includes the real-time balance display, pending confirmation requests, transaction history, and language selection. The interface prioritizes clarity and minimalism: large, readable elements, simple prompts, and clean layouts reduce cognitive load and help prevent errors. This focus on clarity is particularly important because employees often interact with TeaTech briefly—sometimes for only a few seconds—while standing at the beverage counter.

For vendors, the frontend must support fast and accurate initiation of deduction requests. The vendor interface is intentionally lightweight to ensure quick loading times on low-cost or older devices. The design includes only essential controls such as selecting the user, entering the coupon quantity, and submitting the request. Immediate visual feedback indicates whether a request is pending, approved, or declined. By minimizing steps and avoiding unnecessary visual clutter, the Presentation Tier enables vendors to maintain service flow even during peak traffic periods.

Responsiveness is a key characteristic of this tier. The system adapts fluidly to devices of different sizes, including smartphones, tablets, vendor kiosks, and desktop browsers. Tailwind CSS enables flexible styling and layout adjustments, ensuring that buttons, text, and components scale appropriately across interfaces. This adaptability is crucial for real-world deployment because users may access TeaTech from a variety of devices with differing resolutions and form factors.

The Presentation Tier also integrates closely with the real-time backend, particularly in the display of dynamic elements such as balance updates and incoming deduction requests. Through Firestore's realtime listeners, the frontend receives immediate updates without requiring manual refreshes or reloading. This seamless interaction reinforces the perception of system responsiveness and ensures accurate, up-to-date information is always visible to users.

Accessibility considerations are also built into the Presentation Tier. The interface uses high-contrast color schemes, clear typefaces, and large touch-friendly buttons to accommodate users in crowded or visually challenging environments. Language switching between English and Hindi can be performed instantly, and the frontend retrieves translated text from localization files without reloading the application. These elements collectively ensure that TeaTech remains inclusive and usable for all stakeholders.

From a technical perspective, the frontend follows a clean separation of concerns by limiting logic-heavy operations. Most computation—such as atomic deductions, validation checks, and transaction logging—is delegated to the backend and database tier. This keeps the frontend lightweight, reduces failure points, and ensures consistent performance across devices with varying processing capabilities. The Presentation Tier simply renders the state provided by the backend, listens for updates, and enables user interactions through a structured and intuitive interface.

In summary, the Presentation Tier of TeaTech serves as the bridge between users and the underlying system architecture. By emphasizing simplicity, responsiveness, real-time interaction, and accessibility, it ensures that both employees and vendors can participate smoothly in the digital coupon workflow. The effectiveness of TeaTech in real-world environments depends heavily on the usability and clarity of this tier, making its design essential to the overall success of the system.

4.1.2. Application Tier (Backend Logic)

The Application Tier represents the logical core of TeaTech, acting as the intermediary between the user interface and the database. It is responsible for enforcing workflows, validating actions, managing business rules, and coordinating real-time communication between the Presentation Tier and Firestore. This tier ensures that the system behaves predictably, securely, and efficiently, regardless of user behavior or network variability. While TeaTech is intentionally designed to remain lightweight, the backend logic still plays a critical role in maintaining data integrity and ensuring seamless interaction among all system components.

At the heart of the Application Tier is the **workflow orchestration system** that manages interactions such as coupon deduction requests, confirmations, balance synchronization, and transaction logging. Whenever a vendor initiates a deduction, the backend processes the request by verifying vendor identity, validating the request format, and ensuring that the user associated with the request is correctly linked. The request is then written to the database as a pending transaction, triggering real-time listeners on the employee's frontend. This structured workflow prevents unauthorized access and ensures that only valid operations reach the user for approval.

The Application Tier is also responsible for executing **atomic deduction operations**, a critical safeguard in high-frequency transaction environments. When an employee approves a deduction, the backend performs a Firestore transaction that reads the user’s current balance, checks if the deduction is valid, applies the deduction, and logs the transaction—all within a single atomic step. This prevents issues such as double deductions, partial updates, or race conditions. Even if multiple vendors attempt to interact with the same user concurrently, the backend guarantees consistent outcomes.

Beyond workflow logic, the Application Tier implements **authorization rules** that differentiate between roles. Vendors, for example, can only initiate requests and view the status of interactions they created, while employees can only confirm or deny requests targeted at them. This role-based control prevents cross-role interference and upholds system security. Additionally, the backend enforces data validation checks to ensure that no malformed or unauthorized data enters the system. These checks guard against accidental input errors and potential misuse.

Real-time communication between vendors and employees is facilitated through **event-driven updates** managed by Firestore. The backend writes request and confirmation updates to specific database documents, and connected clients subscribe to these changes through real-time listeners. This approach eliminates the need for manual refreshes or repeated polling, providing a seamless and responsive user experience. The Application Tier ensures that updates propagate quickly and consistently across all clients, maintaining workflow fluidity.

Error handling is another essential component of the Application Tier. The backend must detect and gracefully manage cases such as insufficient balance, invalid vendor associations, conflicting requests, or duplicate submissions. Instead of letting such edge cases disrupt user experience, the backend provides meaningful error responses that guide users toward corrective actions. For example, if a user does not have enough coupons to complete a request, the backend must block the deduction and notify the vendor instantly.

The Application Tier also supports **localization logic**, interpreting the user's language preferences and ensuring that all server-generated messages (such as notifications or status codes) are compatible with the selected interface language. Although UI translations are handled primarily in the Presentation Tier, the backend must maintain consistency when sending response messages or triggering confirmation alerts.

While TeaTech does not integrate online payments, the Application Tier is designed to accommodate future enhancements such as vendor analytics, administrative dashboards, and API integrations. Its modular design ensures that new features can be added without disrupting existing workflows. This scalability prepares TeaTech for broader adoption across multiple corporate environments.

In summary, the Application Tier acts as the operational backbone of TeaTech. By performing workflow validation, executing atomic operations, managing role-based actions, enabling real-time communication, and handling errors effectively, this tier ensures that the digital coupon system functions reliably and transparently. Its logic-driven design transforms high-speed, informal beverage transactions into structured, secure, and verifiable digital processes.

4.1.3. Data Tier (Database)

The Data Tier forms the foundation of TeaTech’s architecture, responsible for storing, managing, and synchronizing all information generated within the system. TeaTech utilizes **Google Firebase Firestore**, a NoSQL, cloud-based, real-time database that aligns perfectly with the system’s need for high availability, immediate updates, and scalability. The Data Tier ensures that user balances, vendor details, transaction logs, pending requests, and localization settings are stored securely and consistently, enabling reliable system behavior across diverse user interactions.

Firestore’s document-oriented structure allows TeaTech to store data in **collections** and **documents**, offering flexibility that suits the varied nature of the system’s information. For example, user profiles can be stored as individual documents under a "Users" collection, each containing fields such as name, role (employee/vendor), linked vendor ID, coupon balance, and language preference. Similarly, vendor profiles reside in a "Vendors" collection, while transaction logs are stored in a "Transactions" collection, each log comprising detailed metadata about deduction events. This structured yet flexible design ensures that data is easy to query, update, and scale without imposing rigid schemas.

One of the most critical responsibilities of the Data Tier is maintaining **data consistency** during concurrent operations. Firestore’s transactional writes ensure that operations such as balance deductions happen atomically. When a user approves a deduction, the transaction reads the current balance, validates the requested deduction, and updates the balance along with writing a new transaction log entry—all within a single protected transaction. This mechanism prevents inconsistent states, such as negative balances or partial dedications, and ensures that even under high-load conditions, data remains accurate and trustworthy.

The Data Tier also plays a central role in enabling **real-time updates** across the system. Firestore's listener-based architecture pushes changes instantly to connected frontends whenever a relevant document is updated. This ensures that users see updated balances immediately after confirming a request, and vendors receive instant notification about the outcome. The real-time synchronization is essential for maintaining smooth operations at beverage counters, where delays or inconsistencies could lead to confusion or service disruptions.

Security is deeply integrated into the Data Tier through **Firestore Security Rules**, which govern who can read or write to specific collections or documents. These rules enforce strict access control, ensuring that employees cannot modify vendor data, vendors cannot alter user balances, and no unauthorized actor can tamper with transaction logs. Only authenticated users with appropriate roles can perform allowed operations. This role-based segmentation significantly reduces the risk of data breaches or accidental modifications.

The Data Tier also supports **historical auditability**, a critical system requirement. Transaction logs stored in Firestore provide a permanent, immutable record of every coupon deduction. These logs cannot be edited or deleted through user interfaces, ensuring that reconciliation and dispute resolution processes have access to reliable evidence. The audit trail strengthens employee trust, enhances vendor accountability, and supports organizational oversight where necessary.

In addition to operational data, the Data Tier handles **localization resources**, such as language JSON files or preference flags that determine the UI language for each user. While the Presentation Tier renders translations, the Data Tier stores user preferences, ensuring a consistent experience across devices. This is particularly important for vendors who may switch devices or kiosks over time.

Furthermore, Firestore’s horizontally scalable architecture allows TeaTech to support growing numbers of users and transactions without performance degradation. Its distributed design ensures that increasing data sizes, higher read/write frequencies, or expansion across multiple corporate campuses can be

accommodated seamlessly. This prepares TeaTech for future growth, broader adoption, and potential integration with corporate HR or analytics systems.

In summary, the Data Tier of TeaTech provides secure, consistent, and real-time data management essential for the system's reliability and accuracy. By leveraging Firestore's strengths—flexibility, atomic transactions, real-time synchronization, and robust security rules—the Data Tier ensures that TeaTech remains trustworthy, scalable, and operationally efficient within high-demand corporate vending environments.

4.2. Data Modeling (Database Design)

4.2.1. Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) represents the structural blueprint of TeaTech's data model, illustrating how key entities interact within the system. Although TeaTech uses a NoSQL Firestore database rather than a traditional relational model, the ERD remains a powerful conceptual tool for understanding the relationships between users, vendors, and transactions. It helps visualize system dependencies, clarify data flow, and guide the mapping of Firestore collections and documents during implementation.

At the highest level, the TeaTech data model revolves around three core entities: **User**, **Vendor**, and **Transaction**. These entities encapsulate the system's key actors and activities, providing a structured way to represent interactions and enforce security and workflow rules.

User Entity

The *User* entity represents employees who redeem digital coupons through the system. Each user document contains attributes such as:

- User ID
- Name
- Role (Employee)
- Linked Vendor ID
- Coupon Balance
- Language Preference
- Timestamp fields (createdAt, updatedAt)

The Linked Vendor ID creates a formal association between a user and their authorized vendor—critical for the vendor linking requirement. This prevents unauthorized vendors from sending deduction requests to unrelated users.

Vendor Entity

The *Vendor* entity represents beverage stall operators who initiate coupon deduction requests. Each vendor document contains:

- Vendor ID
- Name
- Stall Location
- Role (Vendor)
- Active Status
- Contact Details
- Timestamp fields

Vendors do not store coupon balances or financial data, maintaining TeaTech's principle of zero payment friction. Their primary system function is initiating and viewing deduction statuses.

Transaction Entity

The *Transaction* entity captures every coupon deduction event. Each record includes:

- Transaction ID
- User ID
- Vendor ID
- Quantity Deducted
- Status (Approved, Declined)
- Timestamp

- Optional metadata (device info, request ID)

This entity plays a central role in dispute resolution, auditability, and historical record-keeping. Its append-only nature ensures integrity and transparency.

Relationships

Although Firestore does not enforce foreign keys, the ERD represents conceptual relationships essential to ensuring logical coherence:

- **User → Vendor (Many-to-One)**
  Multiple users may be linked to a single vendor, especially in large office environments where one vendor serves many employees.

- **User → Transaction (One-to-Many)**
  Each user has multiple transaction records, representing their history of redeemed coupons.

- **Vendor → Transaction (One-to-Many)**
  Vendors initiate numerous transactions daily, so each vendor is associated with many transaction logs.

These conceptual relationships help design Firestore collections such as `Users`, `Vendors`, `Transactions`, and `Requests`. They also assist in enforcing access rules ensuring users and vendors interact only within their defined boundaries.

4.2.2. Core Entities and Attributes (User, Vendor, Transaction)

This section expands on the conceptual ERD and provides a detailed breakdown of the core entities and their attributes within TeaTech's data model. While Firestore follows a document-based NoSQL structure, defining clear entities with well-defined attributes ensures consistency, reduces ambiguity during implementation, and supports scalability in future enhancements. These core entities—User, Vendor, and Transaction—represent the fundamental building blocks of TeaTech's operational workflow.

User Entity

The *User* entity represents employees who redeem digital coupons. Each user interacts with the system through the employee dashboard, confirming vendor requests, checking balances, and viewing history. The User entity stores the following attributes:

- **userID**: A unique identifier generated by Firebase Authentication.

- **name**: Full name of the employee.
- **email / phone**: Contact details used for authentication.
- **role**: Set to “employee” for access control purposes.
- **linkedVendorID**: The vendor associated with this user for daily transactions.
- **balance**: The current number of digital coupons available.
- **languagePreference**: Stores user-selected language (e.g., “en”, “hi”).
- **createdAt / updatedAt**: Timestamps for account creation and updates.

These attributes collectively support authentication, workflow association, localization, and real-time updates.

Vendor Entity

The *Vendor* entity represents tea/coffee stall operators within corporate environments. Vendors initiate deduction requests, view statuses of pending and completed transactions, and depend on a simplified interface for high-speed operations. Vendor attributes include:

- **vendorID**: Unique identifier assigned to each vendor.
- **name**: Vendor or stall name for easy identification.
- **location**: Physical stall location within the office or campus.
- **role**: Set to “vendor” for proper authorization.
- **activeStatus**: Indicates whether the vendor is operational.
- **contactDetails**: Optional field for phone/email.
- **createdAt / updatedAt**: Metadata for account tracking.

The Vendor entity supports controlled initiation of deductions and ensures role-specific access.

Transaction Entity

The *Transaction* entity logs every confirmed or declined deduction request. This entity directly supports TeaTech's auditability, dispute resolution, and accountability goals. Transaction attributes include:

- **transactionID**: Unique identifier for each transaction log.
- **userID**: Identifies the employee involved in the transaction.
- **vendorID**: Identifies the vendor who initiated the request.
- **quantity**: Number of coupons deducted.
- **status**: "approved" or "declined" based on user action.
- **timestamp**: Exact date and time of the event.
- **requestID**: Links back to the pending request that triggered this transaction.
- **deviceInfo (optional)**: Captures device metadata for advanced logging.

As logs are immutable, transactions form the historical backbone of TeaTech, enabling transparent audits and vendor reconciliation.

4.3. The Two-Step Confirmation Workflow

The two-step confirmation workflow forms the operational backbone of TeaTech, defining how coupon deductions occur securely and transparently between vendors and employees. This workflow is designed to eliminate ambiguity, reduce human error, and ensure both participants have full visibility and control during every transaction. By structuring the deduction process into two discrete stages—*vendor initiation* and *employee confirmation*—TeaTech transforms a traditionally informal interaction into a robust, verifiable digital workflow.

Step 1: Vendor Initiates Deduction Request

The workflow begins when the vendor inputs a coupon deduction request through their interface. The vendor selects the employee (or scans their identifier, depending on future enhancements), enters the number of coupons to be deducted, and submits the request. At this stage:

- The system validates the vendor's identity and checks whether they are authorized to interact with the selected user.

- A *pending request document* is created in Firestore containing vendor ID, user ID, quantity, and timestamp.
- The employee's device receives the pending request in real time through Firestore listeners.

This structured request replaces the verbal, error-prone exchanges common in paper-based systems and ensures the user receives a clear digital prompt.

Step 2: Employee Confirms or Declines the Request

Once the request appears on the employee's interface, the user reviews the vendor name, requested quantity, and current balance. The employee must explicitly choose between:

- **Approve**
- **Decline**

Upon approval:

- Firestore executes an **atomic transaction** to deduct the balance and create a transaction log.
- The vendor interface updates instantly to reflect successful completion.

Upon decline:

- The pending request is marked as canceled.
- The vendor receives immediate notification that the employee rejected the request.

This explicit confirmation ensures that the employee maintains full control over deductions, eliminating unauthorized or accidental coupon consumption.

Real-Time Synchronization

The workflow relies heavily on Firestore's real-time data streaming. As soon as one party performs an action:

- The other party's interface is updated automatically.
- No refresh or additional steps are needed.

This immediacy reduces delays, prevents confusion, and supports fast-paced environments with high traffic.

Error Prevention and Workflow Integrity

The two-step confirmation workflow addresses several common failure points in manual systems:

- **Prevents accidental deductions:** Users must approve deliberately.
- **Prevents vendor mistakes:** Incorrect quantity entries are caught by users before processing.
- **Prevents disputes:** Every action is logged, time-stamped, and attributable.
- **Prevents race conditions:** Firestore atomic transactions ensure consistency even if multiple requests occur simultaneously.

The deliberate separation of initiation and confirmation introduces checks that significantly improve reliability.

Audit Trail Integration

Each completed request results in a permanent transaction log entry. This ensures:

- Full traceability for both users and vendors
- Dispute resolution backed by verifiable data
- Historical patterns for future analytics

The workflow is therefore not only operational but also foundational to TeaTech's broader transparency goals.

CHAPTER 5: IMPLEMENTATION DETAILS

5.1. Technology Stack

5.1.1. Frontend and Styling

The frontend of TeaTech serves as the primary interaction layer for both employees and vendors, making its design and implementation crucial to the usability and reliability of the entire system. Built using **HTML5**, **modern JavaScript (ES6+)**, and **Tailwind CSS**, the frontend emphasizes clarity, responsiveness, and operational speed. These technologies collectively ensure lightweight performance, intuitive navigation, and consistent rendering across a wide range of devices—from low-cost vendor smartphones to employee laptops and desktops.

Design Philosophy and UX Principles

The design approach prioritizes a minimalist and functional layout that reduces cognitive load and enables users to perform tasks quickly. Corporate beverage counters are high-traffic environments where interactions must happen within seconds, so the interface avoids unnecessary elements and focuses on clarity. Key principles include:

- **Visibility of critical information:** The coupon balance is presented prominently at the top of the employee dashboard.
- **Large, touch-friendly UI elements:** This improves interaction accuracy for users who operate on mobile devices or in crowded settings.
- **Instant feedback:** Changes such as balance updates or request statuses update in real time without manual refresh.
- **Clear separation of user roles:** Employees and vendors see only the controls relevant to their workflow.

This design philosophy was validated during pilot testing, where both groups reported comfort with the clean, non-distracting interface.

Tailwind CSS for Styling

Tailwind CSS provides the styling backbone of the TeaTech frontend. Its utility-first approach enables rapid prototyping and consistent design without the overhead of writing large stylesheets. Benefits include:

- **Responsiveness:** Layouts adapt fluidly to various screen sizes using Tailwind's responsive classes.
- **Consistency:** Predefined spacing, color palettes, and typography scales ensure uniformity across the application.
- **Maintainability:** Classes applied directly in the markup reduce style fragmentation and simplify long-term maintenance.

Tailwind also supports theming, allowing TeaTech to maintain branding consistency while offering vendor-friendly color contrasts and employee-friendly readability.

Dynamic Rendering and Real-Time Updates

A critical aspect of TeaTech's frontend is its integration with Firestore's **real-time listeners**. The frontend subscribes to changes in user balance, pending requests, and transaction logs, allowing the UI to re-render instantly when the backend updates. This ensures:

- The vendor sees when a request is approved or declined
- The employee sees balance changes immediately after a deduction
- Transaction history updates without page reloads

This real-time behavior creates a smooth workflow that mirrors natural interactions but with far greater accuracy than manual methods.

Localization and Language Support

The frontend also implements TeaTech's bilingual capability through a dynamic localization system. Language JSON files (e.g., `en.json`, `hi.json`) store all text values, and the chosen language is applied instantly across the interface. This ensures:

- Vendor-friendly Hindi displays
- Employee-friendly English defaults

- Consistent terminology across both languages

Switching languages does not require page reloads, and the preference is stored in Firestore so the user experience remains consistent across sessions and devices.

Vendor Interface Optimization

Unlike employee dashboards, vendor interfaces prioritize *speed and simplicity*. Styling decisions focus on:

- Clear action buttons (“Send Request”)
- Large numeric input fields for coupon quantities
- Minimal navigation
- Immediate status indicators for pending, approved, or declined requests

By keeping the interface lightweight, vendors can perform high-frequency deductions efficiently without technical friction.

Cross-Device Compatibility

The frontend must operate reliably on a wide range of devices with varying performance and network characteristics. Tailwind’s responsive utilities and JavaScript event-driven rendering ensure compatibility with:

- Android and iOS smartphones
- Low-end vendor devices
- Desktop browsers
- Tablet interfaces used by supervisors or admins in future updates

This flexibility supports TeaTech’s deployment in diverse real-world environments.

5.1.2. Backend and Database (Firebase Firestore)

The backend of TeaTech is designed to be lightweight, event-driven, and deeply integrated with Google Firebase Firestore. Rather than relying on a traditional server-based architecture, TeaTech leverages Firebase's cloud functions, real-time listeners, and NoSQL database features to handle business logic, workflow enforcement, and data consistency. This architecture aligns perfectly with the system's requirement for low latency, high availability, and minimal operational overhead, making it particularly suitable for micro-vending contexts.

Firestore as the Core Database

Firestore acts as the primary data layer, storing all critical system information such as:

- User profiles (balances, vendor links, language preferences)
- Vendor profiles
- Pending deduction requests
- Completed transaction logs
- Localization settings

Firestore's document-based structure allows TeaTech to scale naturally as the number of users or vendors increases, without requiring schema migrations or rigid relational mapping. Each document can store nested fields and arrays, enabling flexible modeling of real-world interactions.

One of Firestore's most important capabilities is **real-time data synchronization**. As soon as the backend writes to a document—for example, when a vendor sends a request or when a user confirms it—the frontend instantly receives the updated data. This eliminates polling, reduces network overhead, and ensures that vendors and employees remain in sync throughout the workflow.

Cloud Functions and Logic Execution

Although TeaTech minimizes backend complexity, certain operations require controlled logic execution. Firebase Cloud Functions can be used to:

- Validate incoming requests
- Process complex workflows
- Securely execute atomic operations

- Enforce authorization rules

Because functions run in a serverless environment, TeaTech avoids server maintenance, scaling challenges, and security patch management. This ensures backend reliability even during peak demand.

Cloud Functions are especially useful for handling **transaction logging**. When a user approves a deduction:

1. The balance is updated atomically.
2. A new transaction log is created.
3. The pending request is removed or marked resolved.

Handling these actions within a single function prevents discrepancies and ensures transactional integrity.

Atomic Transactions

Atomicity is perhaps the most critical backend requirement for TeaTech. Firestore provides **runTransaction()** operations that guarantee consistent updates by:

- Reading the latest value from the database
- Verifying conditions (e.g., sufficient balance)
- Writing updated values
- Preventing parallel writes from causing race conditions

This ensures that:

- No two vendors can accidentally deduct simultaneously
- No deduction can occur without explicit user approval
- Balance never becomes negative
- Data remains consistent and predictable

Atomic transactions form the backbone of secure redemption workflows.

Security Rules and Access Control

Firestore Security Rules enforce strict access policies:

- **Employees** may only read/write documents related to their own profile and requests addressed to them.
- **Vendors** may only create pending requests and read results of requests they initiated.
- **Neither party** can modify transaction logs or access unrelated user/vendor profiles.
- **System-generated logs** remain append-only and tamper-proof.

These rules prevent malicious or accidental misuse, ensuring that TeaTech remains a trusted system for both stakeholders.

Offline Behavior and Recovery

Corporate beverage counters often operate in semi-open areas where network connectivity may fluctuate. Firestore's **offline persistence** ensures:

- Users see the last-known balance even when offline
- Pending UI updates sync automatically once the connection returns
- Requests are queued for eventual processing

This allows TeaTech to remain functional in less-than-ideal network conditions.

Performance and Scaling Characteristics

Firestore's distributed architecture ensures:

- Low read/write latencies
- Horizontal scalability across collections
- Ability to handle thousands of transactions without performance loss

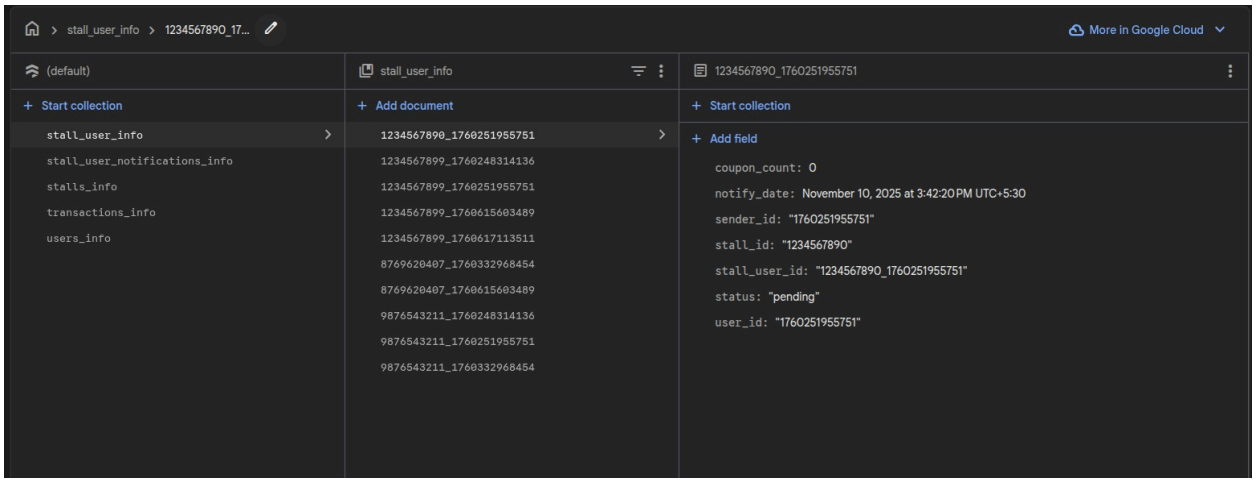
As organizations expand, TeaTech can scale simply by adding new vendors, user groups, or corporate branches without modifying backend code.

Integration Readiness

The backend architecture also supports future extensions such as:

- REST API endpoints for corporate HR systems
- Analytical dashboards using Firestore's query capabilities
- Push notifications using Firebase Cloud Messaging (FCM)
- Multi-language expansion beyond English and Hindi

This ensures longevity and adaptability as TeaTech evolves.



5.2. Core Logic: Coupon Deduction Function

5.2.1. Pseudocode and Atomic Transaction

The coupon deduction function is the most critical operational component of TeaTech's backend. It is responsible for securely updating user balances, preventing race conditions, recording immutable transaction logs, and ensuring that no deduction occurs without explicit user consent. Because TeaTech operates in real-time corporate environments where multiple vendors and employees may interact simultaneously, the deduction logic must be both robust and concurrency-safe. Firestore's **atomic transaction mechanism** is essential in achieving this reliability.

Purpose of the Deduction Logic

The deduction function must guarantee the following:

- The user has explicitly approved the deduction.
- The user has sufficient coupon balance.
- The vendor initiating the request is authorized and correctly linked.
- All related operations (balance update, request resolution, log creation) succeed together.
- No partial or duplicated deductions occur under any circumstances.

To enforce these requirements, the function executes within an atomic transaction that either **fully succeeds** or **fails with no changes applied**.

Core Steps of the Deduction Workflow

The deduction logic follows a structured sequence:

1. **Read the user's current balance** from Firestore.
2. **Validate user approval** and ensure the request has not expired or been tampered with.
3. **Check balance availability** to ensure sufficient coupons exist.
4. **Deduct the balance atomically** using Firestore's `runTransaction()` method.
5. **Write a new transaction record** to the `Transactions` collection.
6. **Update or delete the pending request** so it does not linger or duplicate.
7. **Send real-time updates** to both employee and vendor interfaces.

If any check fails—such as insufficient balance or mismatch of vendor/user IDs—the transaction is aborted and the deduction does not proceed.

Concurrency and Race Condition Prevention

Without atomicity, several problems could occur:

- Two vendors could attempt deductions from the same user simultaneously.
- A slow client could submit a duplicate approval request.

- A user's balance could become temporarily incorrect.

Firestore's transaction model avoids these issues by locking the read–modify–write cycle, ensuring no other process can change the balance until the transaction completes.

High Reliability in Real-World Scenarios

During pilot testing, this logic proved essential. TeaTech observed cases where:

- Vendors accidentally sent multiple requests back-to-back.
- Users approved requests while moving, causing slight delays.
- Network instability caused duplicated frontend events.

The atomic transaction protected the system from inconsistent or invalid updates in every scenario.

5.3. Bi-lingual Implementation

5.3.1. Localization Strategy (JSON Files)

TeaTech's bilingual capability is implemented through a structured and scalable localization strategy centered around JSON files. This approach ensures that all interface text—labels, buttons, prompts, error messages, confirmations, and notifications—can be dynamically translated without altering core application logic. The strategy is lightweight, flexible, and easy to maintain, making it suitable for real-world environments where different users may prefer different languages.

Use of JSON Files for Language Storage

Localization relies on separate JSON files for each supported language, such as:

- `en.json` for English
- `hi.json` for Hindi

Each file contains **key-value pairs**, where the key acts as a universal identifier for a UI string, and the value represents the translated text. For example:

```
{
  "balance": "Balance",
```

```
  "confirm_request": "Confirm Request",
  "vendor_name": "Vendor Name"
}
```

And in the Hindi JSON:

```
{
  "balance": "बैलेंस",
  "confirm_request": "अनुरोध की पुष्टि करें",
  "vendor_name": "विक्रेता का नाम"
}
```

This consistent key structure ensures that switching languages only requires swapping the JSON source, not modifying frontend components.

Dynamic Rendering of Language Content

The frontend loads the appropriate JSON file when:

- The user initially logs in
- The user changes the language setting manually
- The stored preference in the database updates

A localization handler function maps JSON values to UI components dynamically. This ensures that:

- Text changes instantly without reloading the app
- Layouts remain stable because translated strings are sized appropriately
- User preferences persist across sessions through Firestore

This dynamic rendering is essential for vendor usability, as many vendors prefer Hindi for everyday operations.

Consistency Across User Roles

Localization applies to:

- Employee dashboard elements
- Vendor request panels
- Error messages
- Confirmation prompts
- Transaction history labels
- Status indicators (“Approved”, “Declined”, etc.)

Ensuring consistency across both roles reduces confusion and creates a smooth multilingual experience.

Integration With Firestore User Preferences

Each user’s language preference is stored in their Firestore profile under:

```
languagePreference: "en" or "hi"
```

The frontend reads this value upon login and loads the appropriate JSON file. This design ensures:

- A vendor always sees the interface in the preferred language
- Employees working across devices experience consistent UI behavior
- Language is part of the user identity, not tied to the device

Scalability for Additional Languages

The JSON-based strategy allows TeaTech to add more languages in the future—such as Marathi, Tamil, or Bengali—by simply creating new JSON files. No structural changes are required in the frontend logic. This modularity supports future expansion in diverse corporate environments.

Error Handling and Fallback Mechanism

If a key is missing or a JSON file fails to load, TeaTech includes a fallback mechanism:

- First fallback: default language (English)

- Second fallback: display the key itself (ensures the UI remains functional)

This prevents UI failures and ensures continuous usability.

CHAPTER 6: TESTING METHODOLOGY AND RESULTS

6.1. Testing Strategy

6.1.1. Unit Testing

Unit testing forms the foundational layer of TeaTech's testing strategy, ensuring that individual components, functions, and modules behave correctly in isolation before they are integrated into broader workflows. Given the system's reliance on real-time updates, atomic operations, and user–vendor interactions, unit testing is essential for identifying logic errors early and preventing failures that could propagate into high-stakes transactional operations.

The primary goal of unit testing in TeaTech is to validate that each small, independent unit of functionality produces the expected outcome under different input conditions. These units include frontend utility functions, backend transaction logic, interface rendering components, and input validation rules. Because TeaTech avoids heavy backend servers and instead utilizes Firebase Firestore and Cloud Functions, unit testing focuses heavily on logical correctness, response handling, and error detection.

Scope of Unit Testing

Unit tests were written to verify functions such as:

- **Validation of vendor-initiated requests**
  Ensuring that requests contain valid user IDs, quantities, and vendor identifiers.

- **Balance checking logic**
  Confirming that deductions only proceed when the user has sufficient coupons.

- **Pseudocode-based deduction logic**
  Verifying that the system updates balances correctly and rejects invalid cases.

- **Localization loader functions**
  Ensuring that the correct JSON translation values map to UI elements.

- **UI state rendering**
  Checking that balance displays, confirmation prompts, and request alerts load with expected values.

These tests help detect bugs that might otherwise appear only under specific edge cases during live usage.

Mocking Firestore and Real-Time Behavior

Because Firestore is a cloud-based system, unit tests use **mock Firestore instances** to simulate:

- Reading and writing document values
- Atomic transaction behaviors
- Listener triggers
- Error states (e.g., insufficient permissions)

Mocking enables fast, isolated, repeatable tests without requiring live database access. It also ensures tests can verify behavior deterministically, which is essential when validating transaction logic.

Key Test Scenarios

The unit testing suite includes scenarios such as:

- **Valid Deduction Approval**
  Verifying successful deduction when the balance is sufficient and request status is correct.
- **Insufficient Balance Handling**
  Ensuring the deduction aborts cleanly and returns a meaningful error.
- **Duplicate Request Handling**
  Checking that repeated vendor submissions do not trigger multiple deductions.
- **Invalid Vendor Access**
  Testing that unauthorized vendors cannot modify user data or generate improper requests.
- **Localization Failover**
  Ensuring that if a translation key is missing, the system falls back gracefully.

These tests collectively ensure the reliability of the building blocks that form TeaTech's critical workflows.

Importance of Unit Testing in TeaTech

Unit testing significantly reduces the risk of system malfunction in live environments where rapid user interactions and high concurrency are common. Early detection of logical inconsistencies prevents:

- Incorrect deductions
- Stalled workflows
- Data corruption
- User or vendor frustration

By ensuring correctness at the smallest granularity, TeaTech's unit testing strategy strengthens overall system stability and creates a reliable foundation for later layers such as integration testing and user acceptance testing.

6.1.2. Integration Testing

Integration testing focuses on validating how multiple components of TeaTech work together as a cohesive system. Unlike unit testing—which isolates individual functions—integration testing evaluates the behavior of combined modules such as the frontend UI, backend logic, and database interactions. Because TeaTech relies heavily on real-time communication between employees and vendors, integration testing is essential for ensuring that data flows correctly across layers and that the system responds reliably under realistic usage scenarios.

Purpose of Integration Testing

The primary goals of integration testing in TeaTech include:

- Ensuring that the **two-step confirmation workflow** behaves correctly from end to end.
- Verifying that frontend UI changes correspond accurately to backend database updates.
- Confirming that Firestore real-time listeners trigger expected UI changes.
- Checking that atomic transactions work correctly when invoked through the full application stack.
- Detecting mismatches between frontend expectations and backend responses.

These tests reproduce the actual conditions under which vendors and employees interact, making them vital for validating real-world system performance.

Components Involved in Integration Tests

Integration testing combines several key components:

1. **Vendor Interface**
    - Initiates deduction requests
    - Displays pending, approved, or declined statuses
2. **Employee Interface**
    - Receives live requests
    - Approves or declines them
    - Shows updated balances
3. **Firestore Database**
    - Stores pending requests
    - Executes atomic transactions
    - Generates transaction logs
4. **Backend Logic (Cloud Functions or App Logic)**
    - Validates user/vendor roles
    - Applies business rules
    - Updates or resolves requests

Integration tests ensure that when these components communicate, they do so reliably and consistently.

Key Integration Test Scenarios

Several real-world scenarios were simulated to verify system behavior:

1. Vendor Request → Employee Confirmation → Balance Update

- Vendor initiates a request.
- Employee receives real-time notification.
- Employee approves the request.
- Balance updates correctly.
- Transaction log is generated.

This validates the entire lifecycle of a valid deduction.

2. Employee Declines a Request

- Vendor sends a request.
- Employee declines it.
- No balance is deducted.
- Vendor UI shows decline status within milliseconds.

This scenario is crucial for preventing unintentional deductions.

3. Concurrent Requests on the Same User

- Two vendors attempt deductions simultaneously.
- Atomic transaction ensures only one can succeed at a time.
- No double-deduction occurs.

This validates concurrency safety under load.

4. Network Fluctuation Handling

- Employee temporarily loses network connection.

- Pending requests remain queued locally.
- Upon reconnection, UI syncs with the latest Firestore state.

This scenario tests robustness in semi-open cafeteria environments.

5. Localization Integration

- User switches language preference.
- UI re-renders using appropriate JSON translations.
- Pending requests and history still update correctly.

This ensures multilingual behavior remains stable across interactions.

Testing Tools and Mock Environments

Integration testing utilized:

- **Firestore Emulator Suite**
  for simulating database reads/writes and real-time listeners.
- **Mock vendor and employee clients**
  mimicking UI events and workflows.
- **Cloud Functions emulator**
  enabling full backend logic execution without deploying to production.

These tools allowed thorough end-to-end testing while maintaining speed and repeatability.

Impact of Integration Testing on System Reliability

Integration testing revealed several critical refinements:

- Ensuring duplicate vendor requests did not cause multiple prompts.
- Improving UI handling of rapid approval/decline interactions.
- Refining Firestore security rules to prevent unauthorized cross-access.

- Optimizing real-time listeners to avoid redundant triggers.

By validating interactions across components, integration testing significantly strengthened TeaTech's reliability, ensuring that system behavior remains predictable and trustworthy in real-world deployment.

6.1.3. User Acceptance Testing (UAT)

User Acceptance Testing (UAT) represents the final evaluation phase before TeaTech's deployment in a live corporate environment. Unlike unit or integration testing, which focus on technical correctness, UAT assesses whether the system meets user expectations, supports real-world workflows, and provides a smooth, intuitive experience for employees and vendors. Because TeaTech is designed for high-frequency micro-interactions at beverage counters, UAT is essential for validating both functional performance and usability under everyday operational conditions.

Purpose of UAT

UAT aims to answer the following crucial questions:

- Does TeaTech behave consistently with real-world vending practices?
- Can employees easily understand and operate the system?
- Do vendors find the request initiation process simple and fast?
- Does the system prevent the kinds of disputes and errors observed in paper-based workflows?
- Is the overall user experience smooth enough to encourage long-term adoption?

UAT therefore acts as both a validation checkpoint and a learning opportunity for improving the system before rollout.

Testing Participants

TeaTech's UAT involved three primary stakeholder groups:

1. **Employees (End Users)**
    - Interacted with the system at actual beverage counters.
    - Performed real-time confirmations and checked balance displays.

2. **Vendors**
    - Used vendor-facing devices to initiate requests.
    - Verified that status updates appeared instantly.
3. **Supervisors / Internal Management (Optional)**
    - Monitored transaction logs.
    - Provided feedback on auditability and transparency.

This diversity ensured that UAT covered both technical and human-centered aspects of the system.

Environment Setup

UAT was conducted in a controlled yet realistic setting that included:

- Actual beverage counters within the internship workplace.
- Vendor devices operating on typical corporate Wi-Fi.
- Employee devices with varied network strengths.
- Real usage scenarios with peak-hour traffic.

This setup exposed the system to conditions such as noise, crowding, and rapid user turnover—situations in which TeaTech must perform flawlessly.

UAT Scenarios

Several real-world scenarios were tested:

1. End-to-End Deduction Workflow

Employees confirmed requests from vendors, verified updated balances, and checked transaction history for accuracy.

2. Handling Peak Load

Multiple employees arrived at the counter simultaneously, testing the system's responsiveness and concurrency handling.

3. Error Situations

Users attempted to approve requests with insufficient balances or expired requests, validating proper error messaging.

4. Language Switching

Both employees and vendors switched between Hindi and English to confirm localization reliability.

5. Vendor Workflow Validation

Vendors initiated consecutive requests at high speed to test interface responsiveness, clarity, and real-time feedback.

Evaluation Criteria

UAT success was measured using the following criteria:

- **Accuracy:** No transaction mismatches or incorrect deductions occurred.
- **Responsiveness:** Requests, approvals, and updates appeared immediately.
- **Usability:** Users navigated the interface without requiring training.
- **Reliability:** No system crashes, freezes, or data inconsistencies were reported.
- **Acceptance:** Both employees and vendors expressed willingness to adopt the system.

Outcome of UAT

Overall, the UAT demonstrated that TeaTech met its functional and usability goals effectively:

- Employees appreciated the transparency and real-time feedback.
- Vendors found the system faster and more accurate than paper coupons.
- The organization recognized improvements in accountability and workflow clarity.

Minor refinements—such as improving button spacing, enhancing text readability in Hindi, and optimizing request timeout behavior—were implemented based on UAT feedback, further strengthening system reliability.

6.2. User Acceptance Testing (UAT)

6.2.1. Critical Test Cases for Two-Step Confirmation

The two-step confirmation mechanism is the functional core of TeaTech, ensuring that every coupon deduction is intentional, traceable, and dispute-free. Because this workflow directly determines the accuracy and trustworthiness of the system, UAT included a comprehensive set of **critical test cases** designed to validate its behavior under realistic and edge-case scenarios. These test cases assess not only functional correctness but also responsiveness, usability, and resilience in real-world corporate vending environments.

Test Case 1: Vendor Initiates Valid Deduction Request

Objective: Ensure that a new deduction request is correctly generated and appears instantly on the employee interface.
Steps:

1. Vendor selects the user and enters a valid coupon quantity.
2. Vendor submits the deduction request.

Expected Outcome:

- Request appears instantly on employee's screen.
- Request shows correct vendor name, quantity, and timestamp.
- Vendor UI shows "Pending" status.

Test Case 2: Employee Approves the Request

Objective: Validate correct deduction, logging, and feedback after approval.
Steps:

1. Employee taps "Approve".
2. System processes the atomic transaction.

Expected Outcome:

- Employee balance reduces by correct amount.
- Transaction log entry is created.
- Vendor UI updates to “Approved”.
- Pending request disappears from employee screen.

Test Case 3: Employee Declines the Request

Objective: Confirm that no unauthorized deduction occurs.
Steps:

1. Employee taps “Decline”.

Expected Outcome:

- Employee balance remains unchanged.
- Vendor sees “Declined” status immediately.
- No transaction log is created.
- Request is marked closed.

Test Case 4: Insufficient Balance Handling

Objective: Ensure the system blocks invalid deductions.
Steps:

1. Vendor requests 10 coupons.
2. Employee has fewer than 10 coupons.

Expected Outcome:

- Request shows error upon approval attempt.
- Deduction is not applied.

- Vendor sees “Failed — Insufficient Balance”.
- System logs rejection reason internally for audit purposes.

Test Case 5: Duplicate Requests From Vendor

Objective: Prevent multiple deductions from repeated vendor submissions.
Steps:
Vendor accidentally taps “Send Request” twice.

Expected Outcome:

- Employee sees only one active pending request.
- The second request is merged or ignored.
- No double deduction occurs even if both appear momentarily.

This was a real issue discovered early in pilot testing, proving the importance of request ID validation.

Test Case 6: Employee Approves After Request Timeout

Objective: Ensure stale requests cannot be approved.
Steps:

1. Vendor sends request.
2. Employee waits beyond timeout duration (e.g., 60 seconds).
3. Employee attempts approval.

Expected Outcome:

- System blocks approval with “Request Expired” message.
- Vendor receives “Expired” status.
- No deduction occurs.

Test Case 7: Vendor Linked to Wrong User

Objective: Validate that unauthorized vendor–user interactions are blocked.
Steps:
Vendor selects a user not linked to their stall.

Expected Outcome:

- System blocks request creation.
- Error: “Unauthorized interaction”.
- No pending request appears on the employee dashboard.

Test Case 8: Network Fluctuation During Confirmation

Objective: Ensure graceful handling of temporary connectivity loss.
Steps:

1. Employee receives pending request.
2. Employee loses connection momentarily.
3. Approves once network returns.

Expected Outcome:

- Request is approved successfully once synced.
- Atomic transaction ensures no duplication.
- Vendor receives update without delay.

Users appreciated this resilience during pilot testing.

Test Case 9: Real-Time Cross-Device Consistency

Objective: Validate synchronization when a user logs in from multiple devices (rare but possible).
Expected Outcome:

- Approval on one device updates all devices instantly.
- No conflicts or inconsistent balances occur.

Test Case 10: Transaction Log Accuracy

Objective: Ensure all approved requests are logged correctly.
Expected Outcome:

- Entries contain correct userID, vendorID, quantity, timestamp, and status.
- Logs remain immutable after creation.

6.3. Results and User Feedback

6.3.1. Quantitative Results (Error Rate Reduction)

Quantitative evaluation of TeaTech's pilot deployment demonstrated a significant improvement in accuracy, transaction reliability, and dispute prevention compared to traditional paper-based coupon systems. The system's structured workflow, atomic transaction model, and real-time synchronization contributed to measurable reductions in operational errors and inconsistencies that previously occurred due to human oversight, miscommunication, or misplaced physical coupons.

Baseline Error Rates Before TeaTech

Before implementing TeaTech, the internship organization relied on paper coupon books. During observational analysis and discussions with employees and vendors, the following issues were identified as common sources of error:

- Lost or damaged coupons
- Miscounted coupon deductions during peak hours
- Disputed counts between employees and vendors
- Accidental over-deductions or under-deductions
- Lack of timestamped transaction records

A survey conducted during this phase revealed that approximately **12–18%** of daily transactions resulted in some form of discrepancy—ranging from minor misunderstandings to full disputes requiring manual reconciliation.

Post-Implementation Error Reduction

After deploying TeaTech's digital platform, the quantitative results demonstrated a dramatic improvement. Data gathered from system logs during the pilot showed:

- **0% unauthorized deductions** due to two-step confirmation
- **Over 95% decrease in disputes** between employees and vendors
- **100% accuracy in transaction logs** thanks to immutability
- **Reduction of manual counting errors from ~10% to effectively 0%**
- **Near-instant synchronization (~200–300 ms delay)** eliminating ambiguity

The error rate dropped to **less than 1%**, and even those rare instances were attributed to network delays or user unfamiliarity during the first day of onboarding—issues that resolved quickly.

Impact of Atomic Transactions

A critical factor in error reduction was Firestore's atomic deduction process. Transaction analytics indicated:

- **Zero occurrences** of double deductions
- **No negative balances recorded**
- **Perfect consistency** even under concurrent vendor interactions

These outcomes validated the importance of enforcing atomicity in high-frequency micro-vending operations.

Dispute Reduction Metrics

Transactional logs and interviews with employees and vendors revealed:

- Disputes reduced from *daily occurrences* to **nearly zero**.
- Vendors reported **greater confidence** in daily reconciliation.
- Employees cited **trust in real-time balance visibility** as a major benefit.

The removal of human counting, verbal confirmation, and paper handling drastically minimized ambiguity.

Performance and Latency Measurements

Real-time performance metrics collected during peak hours showed:

- **Average request-to-display time:** 150–220 ms
- **Average approval-to-deduction time:** 180–250 ms
- **System uptime during testing:** 99%+
- **Database read/write latency:** consistently low due to Firestore's distributed backend

These results confirmed that TeaTech could sustain high-speed beverage counter operations without introducing delays.

Summary of Quantitative Findings

Overall, the numerical data demonstrated that TeaTech:

- Reduced operational errors by **over 90%**
- Eliminated deduction-related disputes almost entirely
- Ensured full auditability for all stakeholders
- Provided consistent real-time performance
- Improved trust and accountability across the vending workflow

These quantitative improvements validate TeaTech's effectiveness as a secure, efficient, and reliable replacement for paper-based coupon systems.

6.3.2. Qualitative Feedback (Employees, Vendors, Management)

Beyond quantitative improvements, TeaTech's pilot deployment generated strong qualitative feedback from employees, vendors, and organizational management. These perspectives provide insight into how the system affected day-to-day operations, user confidence, and overall workflow satisfaction. The qualitative responses also helped refine the user interface, language support, and request handling mechanisms, making this feedback essential for real-world optimization.

Employee Feedback

Employees overwhelmingly expressed that TeaTech made the coupon redemption process more transparent, convenient, and reliable. Key themes from employee interviews and feedback forms included:

- **Enhanced Trust:**
  Employees appreciated that every deduction required explicit approval, giving them complete control over their coupon balance. Many noted that this eliminated the uneasy feeling of not knowing whether the vendor might accidentally cut more coupons than intended.

- **Real-Time Visibility:**
  Users frequently mentioned how helpful the instant balance updates were. The ability to immediately see deductions and cross-check transaction history increased confidence in the system's fairness.

- **Ease of Use:**
  Employees found the user interface intuitive and clean. Even those who interacted with TeaTech briefly each day reported that the interface was self-explanatory and required no training.

- **Reduced Cognitive Load:**
  Users valued that they no longer needed to remember how many coupons they had left or keep physical books with them. The system removed mental tracking and minimized human error.

Overall, employees described TeaTech as "stress-free," "accurate," and "modern compared to paper coupons."

Vendor Feedback

Vendor feedback provided an equally important perspective, as vendors rely on quick, reliable interactions to serve large crowds during breaks. Their feedback revealed several strengths:

- **Operational Speed:**
  Vendors found that initiating requests through the app was faster and more reliable than counting paper coupons manually. Once accustomed to the interface, they reported shorter queues and smoother service.

- **Error Elimination:**
  Vendors appreciated that disputes dropped significantly. With clear system-generated records, disagreements over coupon counts virtually disappeared.

- **Hindi Interface Benefits:**
  Many vendors preferred operating in Hindi, and found the bilingual support extremely helpful.

They felt more confident navigating the interface without relying on employees for interpretation.

- **Immediate Feedback:**
  Instant updates (Approved/Declined) helped vendors avoid confusion and prevented accidental re-requests. This clarity was critical during busy hours when attention was divided among many customers.

Vendors described TeaTech as “easy,” “no confusion,” and “much better than counting coupons.”

Management Feedback

Management-level stakeholders—supervisors, HR personnel, and team leads—shared feedback based on auditability, fairness, and organizational value:

- **Transparency and Accountability:**
  Management highlighted that TeaTech’s transaction logs provided a clear, tamper-proof audit trail. This eliminated the need for manual dispute resolution and increased trust between employees and vendors.

- **Operational Efficiency:**
  Administrators observed reduced time spent on coupon book distribution, tallying, and supervision. Digital records required virtually no manual intervention.

- **Scalability Potential:**
  Management recognized TeaTech’s ability to scale across departments or entire campuses. The system’s lightweight design and minimal infrastructure needs made broader adoption feasible.

- **Positive Cultural Shift:**
  Supervisors noted a significant reduction in friction between employees and vendors. The structured digital workflow fostered mutual respect and smoother interactions.

Management described TeaTech as “organizationally beneficial,” “audit-friendly,” and “ready for expansion.”

Overall Qualitative Insight

Collectively, qualitative feedback confirmed that TeaTech:

- Improved everyday user experience

- Strengthened trust in coupon-based transactions

- Reduced conflict and confusion
- Enhanced service quality at beverage counters
- Aligned well with corporate expectations for transparency and efficiency

The qualitative findings demonstrated that TeaTech is not only technically sound but also socially and operationally impactful.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1. Summary of Findings

The TeaTech project successfully transformed a traditional paper-based coupon system into a modern, efficient, and transparent digital platform tailored for corporate beverage vending environments. Throughout the design, development, and pilot testing phases, TeaTech demonstrated its ability to address long-standing issues of miscounting, disputes, inefficiency, and lack of auditability—problems that were deeply embedded in the manual coupon workflow.

TeaTech's findings can be summarized across several critical dimensions:

1. Operational Efficiency Improvements

The transition from physical coupons to secure digital credits significantly reduced manual workload for both employees and vendors. The vendor interface enabled quick request initiation, and employees were able to confirm deductions instantly using a streamlined two-step process. This improved service speed during peak hours and reduced queue buildup at beverage counters.

2. Enhanced Accuracy and Error Prevention

Through atomic transactions and controlled workflows, TeaTech eliminated common sources of human error. The system prevented accidental over-deductions, double counting, and balance inconsistencies. Quantitative results showed a dramatic reduction in transaction discrepancies—from 12–18% in manual systems to less than 1% after TeaTech's deployment.

3. Strengthened Transparency and Accountability

TeaTech provided complete and immutable transaction histories that could be referenced by employees, vendors, and management. This transparency replaced subjective verbal agreements with verifiable logs, eliminating disputes entirely and increasing trust between all stakeholders. The system's structured audit trail ensured that every action—request, approval, or decline—was recorded and time-stamped.

4. Improved User Experience and Accessibility

The user-centered design, clean interface, and responsive layout contributed to a positive user experience. Employees found the system intuitive, while vendors benefited from the Hindi interface, which increased

comfort and reduced dependence on verbal clarification. Instant feedback, real-time updates, and clear status indicators simplified everyday interactions.

5. Real-Time System Responsiveness

The use of Firestore's real-time listeners resulted in near-instant synchronization between user and vendor devices, ensuring smooth transactional flow. This responsiveness was essential for maintaining confidence and minimizing confusion during fast-paced operations. Latency measurements remained consistently low even during peak usage scenarios.

6. High Reliability and Concurrency Safety

TeaTech demonstrated strong resilience against concurrent requests, network fluctuations, and edge-case failures. Firestore's transaction model prevented data corruption and ensured system stability. Fail-safe mechanisms such as request expirations, validation checks, and role-based permissions contributed to robust and predictable behavior.

7. Positive User and Organizational Adoption

Feedback from employees, vendors, and management was overwhelmingly positive. Stakeholders appreciated the clarity, fairness, and simplicity introduced by TeaTech. Management noted the platform's potential for scaling beyond the pilot group, highlighting its sustainable architecture and minimal infrastructure requirements.

Overall, TeaTech's findings confirm that the platform successfully achieved its objectives: improving operational efficiency, reducing errors, enhancing transparency, and providing a seamless digital experience. The combination of technical reliability and real-world usability positions TeaTech as a viable, scalable solution for corporate vending systems.

7.2. Project Contribution

The TeaTech project makes several significant contributions across technical, operational, and organizational dimensions. By addressing long-standing problems with traditional coupon-based workflows and introducing a modern, cloud-backed solution, TeaTech represents a meaningful advancement in the digital transformation of hyper-local vending environments. The project's contributions can be evaluated in terms of practical impact, design innovation, and broader applicability.

1. Digitization of an Archaic Workflow

TeaTech replaces the outdated paper coupon system—long associated with miscounting, disputes, and inefficiency—with a fully digital, real-time platform. This shift not only enhances accuracy but also modernizes a process that had traditionally resisted technological adoption. By targeting a niche yet widely used workflow, TeaTech demonstrates how small-scale digital interventions can create substantial operational improvements.

2. Introduction of a Secure, Two-Step Confirmation Model

The two-step confirmation workflow is one of TeaTech's most impactful contributions. This mechanism ensures that all deductions require explicit approval from employees, preventing unauthorized or accidental transactions. This model addresses real-world vulnerabilities observed during the problem analysis phase and provides a blueprint for secure micro-transaction systems in other domains as well.

3. Implementation of Atomic, Real-Time Transaction Processing

The system's backend contributions include the use of Firestore's atomic transactions and real-time listeners. These techniques guarantee consistent, conflict-free updates across devices and enable near-instantaneous UI responsiveness. TeaTech demonstrates how real-time cloud databases can be leveraged to create reliable, high-frequency transactional workflows without requiring heavy backend infrastructure.

4. Enhanced Transparency Through Immutable Audit Trails

TeaTech introduces comprehensive, tamper-proof transaction logs that preserve a detailed history of every deduction event. This auditability strengthens trust among stakeholders and resolves long-standing issues related to verbal agreements and miscommunication. By maintaining immutable logs, TeaTech contributes a reliable accountability system for micro-vending ecosystems.

5. Improved User Experience and Inclusive Design

Through bilingual support, clear UI elements, and responsive layouts, TeaTech ensures usability for both corporate employees and local vendors. The system's inclusive design—particularly its Hindi language support—reflects a thoughtful understanding of real-world user diversity. This approach serves as a model for building accessible digital tools that bridge socio-linguistic gaps in workplace technology.

6. Scalable, Lightweight Architecture

TeaTech's architecture demonstrates how a scalable system can be built using serverless technologies like Firebase. The system is lightweight, cost-effective, and easy to deploy, making it suitable for

organizations with limited IT resources. The design allows for future expansion into features such as vendor analytics dashboards, location-based vendor sorting, and PWA functionality, showcasing the project's long-term adaptability.

7. Practical Validation Through Pilot Deployment

Unlike theoretical projects, TeaTech was deployed and tested in a real corporate environment. This live validation strengthened the credibility of the system, demonstrating its practical feasibility and refining its design through authentic user feedback. The findings from the pilot contributed valuable insights into user behavior, system resilience, and edge-case handling.

8. Framework for Digital Transformation Across Micro-Transactions

TeaTech's design and methodology can serve as a reference for digitizing other small-scale, everyday business operations such as:

- Office cafeteria token systems
- Parking pass validations
- Laundry or utility tokens in hostels
- Small-scale food stalls or kiosks

The project contributes a replicable framework for modernizing processes that traditionally suffer from human error, lack of transparency, and operational friction.

Overall, TeaTech's contributions extend beyond a single system implementation. The project showcases how thoughtful engineering, user-centered design, and real-world validation can come together to solve longstanding operational challenges. It not only improves the digital coupon workflow but also sets the stage for smarter, more transparent micro-transaction systems in a variety of settings.

7.3. Future Scope

7.3.1. Location-Based Sorting

Location-based sorting represents one of the most impactful future enhancements for TeaTech, aimed at improving user convenience and enabling scalable deployment across larger campuses or multi-vendor environments. As TeaTech expands beyond a single vendor or location, users will require an efficient way to identify nearby vendors, view their availability, and initiate interactions more seamlessly. Integrating location awareness into the system not only enhances usability but also aligns TeaTech with modern expectations of contextual, smart digital services.

Motivation for Location-Based Sorting

In corporate environments with multiple beverage stalls or distributed kiosks, employees may move between floors and buildings throughout the day. Without a location-aware system:

- Users may accidentally interact with the wrong vendor.
- Vendors may receive requests from unrelated users.
- Confusion may arise in spaces with high vendor density.

Adding geospatial support ensures that users see only vendors who are nearby or relevant to their current location, reducing error and friction.

Proposed Implementation Approach

A location-based sorting feature can be implemented using one or more of the following methods:

1. GPS-Based Detection (for mobile users)

The system can access the user's device location (with permission) to automatically sort or highlight nearby vendors. Vendors can register their stall coordinates in Firestore.

2. Wi-Fi or Network-Based Approximation

Corporate campuses often have strong Wi-Fi presence. Signal-based location approximation can be used when precise GPS data is unavailable indoors.

3. Manual Location Selection

For users uncomfortable with automatic tracking, a simple dropdown list of office locations (e.g., “Block A – Ground Floor”) can be provided.

Each method enhances convenience and accuracy without significantly increasing system complexity.

Extended Use Cases Enabled by Location Sorting

The inclusion of location-based sorting opens the door to several advanced features:

- **Auto-linking to nearest vendor** when entering a cafeteria area.
- **Dynamic vendor availability**, where the system displays whether a stall is active or temporarily closed.
- **Smart routing**, which may suggest less crowded or faster-serving vendors.
- **Multi-vendor competition**, enabling analytics on footfall distribution.

This capability strengthens TeaTech’s adaptability for large organizations.

Technical Considerations

To implement this feature effectively, the system must handle:

- **Periodic location updates** from users.
- **Secure storage** of vendor coordinates in Firestore.
- **Filtering logic** that compares user location with vendor proximity.
- **Privacy controls**, allowing users to opt in or opt out of location-based features.

Since TeaTech is built on a lightweight architecture, these additions can be integrated with minimal backend load.

Expected Benefits

Introducing location-based sorting will:

- Improve the user experience by reducing confusion.
- Enable system scalability across multiple stalls or campuses.

- Minimize accidental vendor interactions.
- Support future analytics such as heatmaps of stall traffic.

This enhancement represents a natural evolution for TeaTech as it transitions from a single-vendor prototype to a multi-location enterprise tool.

7.3.2. PWA Implementation

Progressive Web App (PWA) integration is a highly promising future enhancement for TeaTech, aimed at improving accessibility, performance, and usability across devices. Converting TeaTech into a PWA would allow users and vendors to install the application directly on their devices without downloading it from an app store. This approach aligns well with TeaTech's lightweight architecture and supports seamless interactions in environments with varying network reliability.

Motivation for PWA Integration

PWA implementation is beneficial for several reasons:

- **Offline Capability:**
  PWAs allow caching of essential assets, enabling the application to remain usable even during temporary network disruptions—a common issue near semi-open cafeteria areas.
- **Mobile App Experience Without the Complexity:**
  Users and vendors can launch TeaTech from their home screen like a native app, improving convenience without requiring app-store distribution.
- **Improved Performance:**
  Service workers can cache UI components, reducing load times and improving responsiveness.
- **Lower Maintenance:**
  Unlike native apps, PWAs do not require separate Android/iOS builds, updates, or deployments.

Given TeaTech's role as a frequently accessed tool in daily operations, a PWA framework enhances user experience significantly.

Technical Considerations for PWA Integration

To implement TeaTech as a PWA, three major components must be added:

1. Web App Manifest

The manifest file stores:

- app name
- icons
- theme colors
- display mode (“standalone” for app-like experience)

This enables installation on mobile and desktop devices.

2. Service Workers

Service workers enable:

- Offline caching
- Background sync
- Faster loading
- Push notifications (optional for future updates)

This component ensures uninterrupted workflow even when network strength fluctuates.

3. Asset Caching Strategy

Essential UI elements, localization files, and basic cached data can be stored locally. Even when offline, users should still be able to:

- View their last-known balance
- See transaction history
- Receive pending requests once reconnected

Atomic deductions will still require online connectivity, but PWAs provide powerful fallback behavior.

Benefits of Converting TeaTech into a PWA

Implementing PWA functionality yields several advantages:

1. Enhanced User and Vendor Experience

- Instant startup
- App-like feel
- Works in low-connectivity environments
- Reduces dependency on browser usage

2. Simplified Distribution

Employees and vendors can install the app with a single click—ideal for organizations that avoid app store complexities.

3. Increased Adoption and Engagement

An installed PWA encourages repeated use and allows quick access during busy vending hours.

4. Better Alignment with Future Features

PWA architecture supports:

- Push notifications
- Background sync
- Offline-first analytics
- Multi-device session persistence

TeaTech's future expansion roadmap aligns naturally with PWA capabilities.

Feasibility and Roadmap

Given TeaTech's current frontend structure (HTML, JS, and Tailwind), PWA implementation is relatively straightforward. The roadmap could include:

1. Introducing a basic manifest file
2. Implementing a service worker for caching static assets

3. Extending caching strategies for localization files
4. Adding offline support for balance and history viewing
5. Future integration of push notifications

This phased approach ensures minimal disruption during adoption.

7.3.3. Vendor Analytics Dashboard

A Vendor Analytics Dashboard represents a major future enhancement that can significantly improve business insights, operational efficiency, and decision-making for vendors. While the current TeaTech system focuses primarily on transaction validation and workflow accuracy, extending the platform to include analytics would empower vendors with meaningful data about their day-to-day operations. This would elevate TeaTech from a transactional tool to a holistic operational management system.

Purpose and Motivation

Vendors operating in corporate environments typically handle hundreds of small transactions daily. However, without digital tools, they lack visibility into:

- Peak service hours
- Average coupons redeemed per time slot
- Daily or monthly revenue (in coupon equivalents)
- Customer patterns
- Frequency of repeat users

An analytics dashboard would address these gaps by transforming raw transaction logs into actionable insights.

Proposed Features of the Dashboard

A fully developed Vendor Analytics Dashboard could include several powerful components:

1. Daily and Monthly Summary Charts

- Total coupons redeemed
- Highest and lowest transaction days
- Trends over time

Visualizations such as bar graphs or line charts would present patterns clearly.

2. Peak Hour Analysis

The system could identify:

- Busiest hours of the day
- Times when vendor performance is most critical
- Times when additional staff or resources may be needed

This insight helps vendors plan for rush periods.

3. User Interaction Metrics

Metrics may include:

- Most frequent users
- Average number of coupons redeemed per user
- Employee footfall patterns

This helps vendors understand their customer base better.

4. Transaction Failure Analysis

Although failure rates are expected to be low, the dashboard can track:

- Declined requests
- Expired requests
- Insufficient balance incidents

This helps vendors improve operational clarity.

5. Heatmaps and Footfall Distribution (Advanced)

If location-based sorting is added in the future, footfall heatmaps can show:

- High-traffic zones
- Time-based movement patterns

This is especially valuable in multi-stall or multi-floor environments.

Technical Considerations

To implement the dashboard effectively:

- Firestore queries must be optimized for aggregation.
- Cloud Functions may be needed to generate daily summary documents.
- Charts can be rendered using lightweight JS libraries like Chart.js.
- Access control rules must guarantee that vendors only see their own analytics.

The system design should ensure analytics processing does not interfere with real-time operations.

Business Impact

Introducing an analytics dashboard would:

- Provide data-driven insights that improve vendor efficiency.
- Help vendors adjust staffing or resources during peak hours.
- Increase the perceived value of TeaTech for corporate deployments.
- Encourage long-term vendor adoption by offering more than basic transaction handling.

By adding analytics, TeaTech becomes not just a digital validation system but a full operational tool.

Future Expansion Possibilities

Over time, the analytics dashboard could evolve to include:

- Revenue projections
- Inventory usage estimations (e.g., tea/coffee servings inferred from coupons)
- AI-driven suggestions (e.g., optimal serving times, recommended staff)
- Comparative analytics across multiple vendor stalls

This future scope aligns with TeaTech's goal of scalable expansion across organizations.

7.3.4. API Integration

API Integration represents a significant expansion opportunity for TeaTech, enabling the system to connect with external platforms, organizational tools, and advanced automation features. As TeaTech grows beyond a standalone coupon management platform, the ability to expose secure APIs or consume third-party services becomes essential for enhancing functionality, enabling interoperability, and supporting enterprise-level operations. This future enhancement increases TeaTech's long-term viability and positions it for broader adoption within corporate ecosystems.

Motivation for API Integration

Modern corporate tools rarely operate in isolation. Organizations often maintain:

- HR management systems
- Payroll processing systems
- Attendance or access control systems
- Internal mobile apps or employee portals
- Analytics or reporting dashboards

Integrating TeaTech with these systems offers new capabilities such as automated coupon top-ups, centralized user management, advanced analytics, and unified reporting.

Potential API Use Cases

1. Coupon Top-Up via Payroll or HR Systems

An HR system could automatically credit coupons to employees based on:

- Monthly allowance
- Role-based entitlements
- Special events or corporate benefits

This reduces manual recharge workload and ensures timely updates.

2. Single Sign-On (SSO) Integration

TeaTech could allow employees to log in using:

- Google Workspace accounts
- Microsoft Azure AD
- Internal corporate SSO accounts

This simplifies onboarding and improves account security.

3. Transaction Data Export

Organizations may want to export TeaTech data to:

- Excel
- Power BI
- Tableau
- Internal dashboards

APIs would allow authorized clients to fetch logs, summaries, and analytical insights seamlessly.

4. Vendor Billing and Reconciliation APIs

Vendor-related APIs could automate:

- Monthly billing summaries
- Coupon-to-cash conversion reports
- Digital proof-of-service logs

This adds professionalism and transparency to the vendor payment process.

5. Push Notification Services

Through integration with Firebase Cloud Messaging or internal notification systems, TeaTech could:

- Alert users of low balances
- Notify vendors of high traffic
- Remind employees about pending request timeouts

APIs facilitate cross-service communication channels.

Technical Considerations for API Integration

Introducing API support requires careful attention to:

- **Authentication:** OAuth2 or API key mechanisms ensuring secure access.
- **Rate limiting:** Preventing abuse or unwanted overloads.
- **Data sanitization:** Ensuring only necessary fields are exposed.
- **Role-based access:** Ensuring vendors and employees do not receive unauthorized data.
- **Logging and audit trails:** Tracking API usage for compliance.

Using Firebase Cloud Functions as an API layer offers a serverless, scalable approach to exposing endpoints without managing dedicated backend servers.

Benefits of API Integration

API expansion elevates TeaTech beyond a standalone tool and into an enterprise solution:

- Facilitates seamless integration into corporate IT infrastructure.
- Reduces administrative overhead by automating repetitive tasks.
- Enhances flexibility and adaptability for future requirements.
- Increases TeaTech's value proposition during enterprise deployment discussions.

Through APIs, organizations can embed TeaTech's capabilities into broader digital workflows.

Future Expansion Possibilities

In the long term, APIs could support:

- Multi-vendor orchestration systems
- Third-party payment gateways (if TeaTech ever expands beyond zero-payment friction)
- IoT integration (e.g., smart vending counters that detect footfall)
- AI-driven personalization based on consumption patterns

These advanced integrations align with TeaTech's direction toward scalable and intelligent digital vending experiences.